

ML Hybrid Models Comparison Report

Generated: 2025-11-24T22:09:18.699589

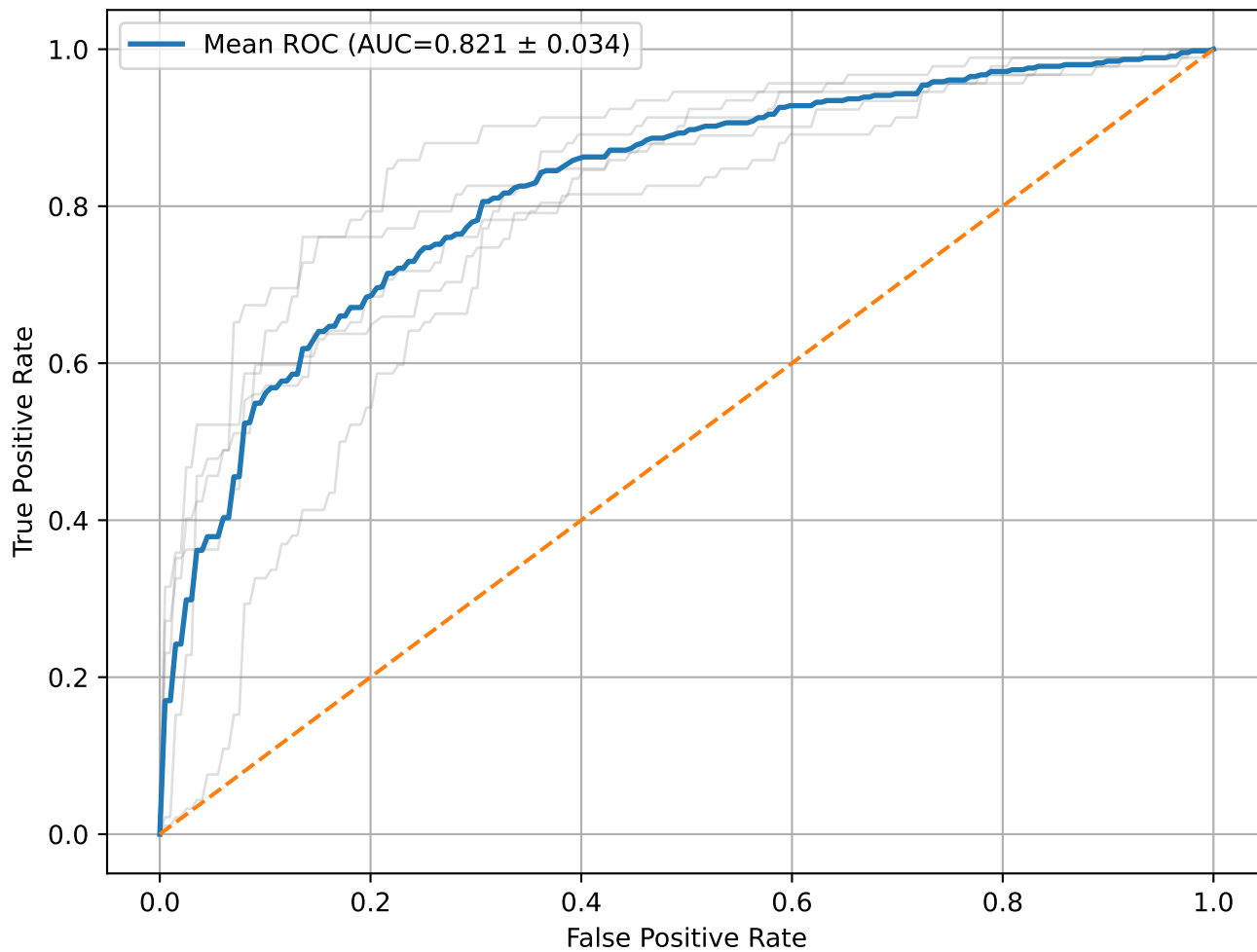
Dataset samples: 906, features: 247

Models: logistic, rf, svm, xgb, lgb, dnn, stacking, voting, bagging, pca_svm, ga_xgb, pso_svm, bayes_opt_svms, gbd

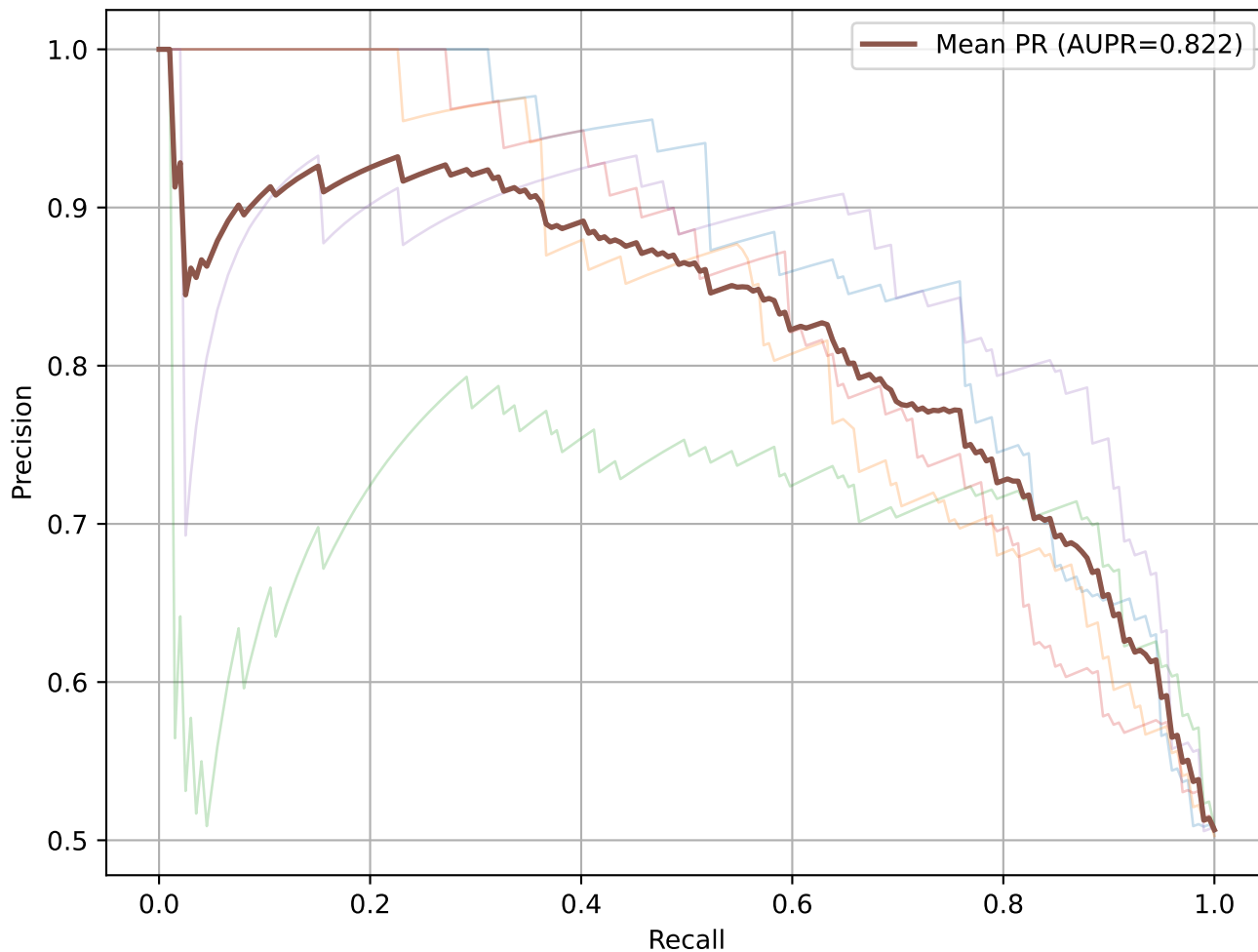
StratifiedKFold: 5 splits

Random seed: 42

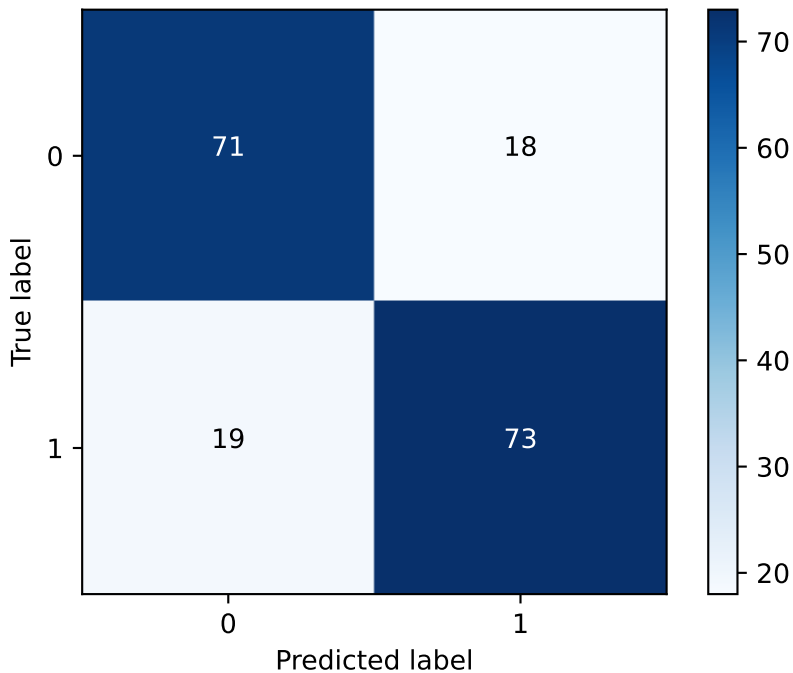
ROC - LOGISTIC



Precision-Recall - LOGISTIC



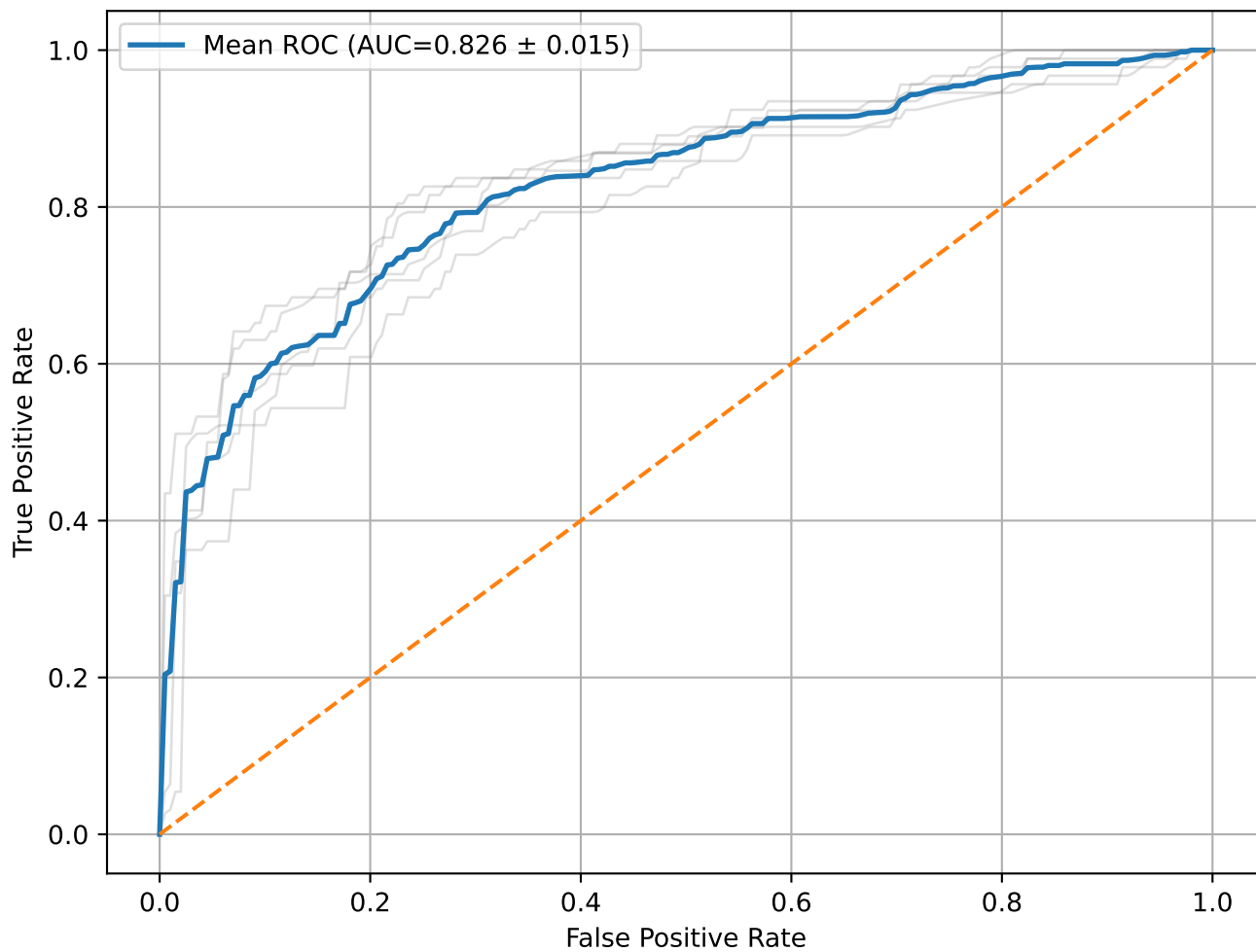
Confusion Matrix - LOGISTIC (last fold)



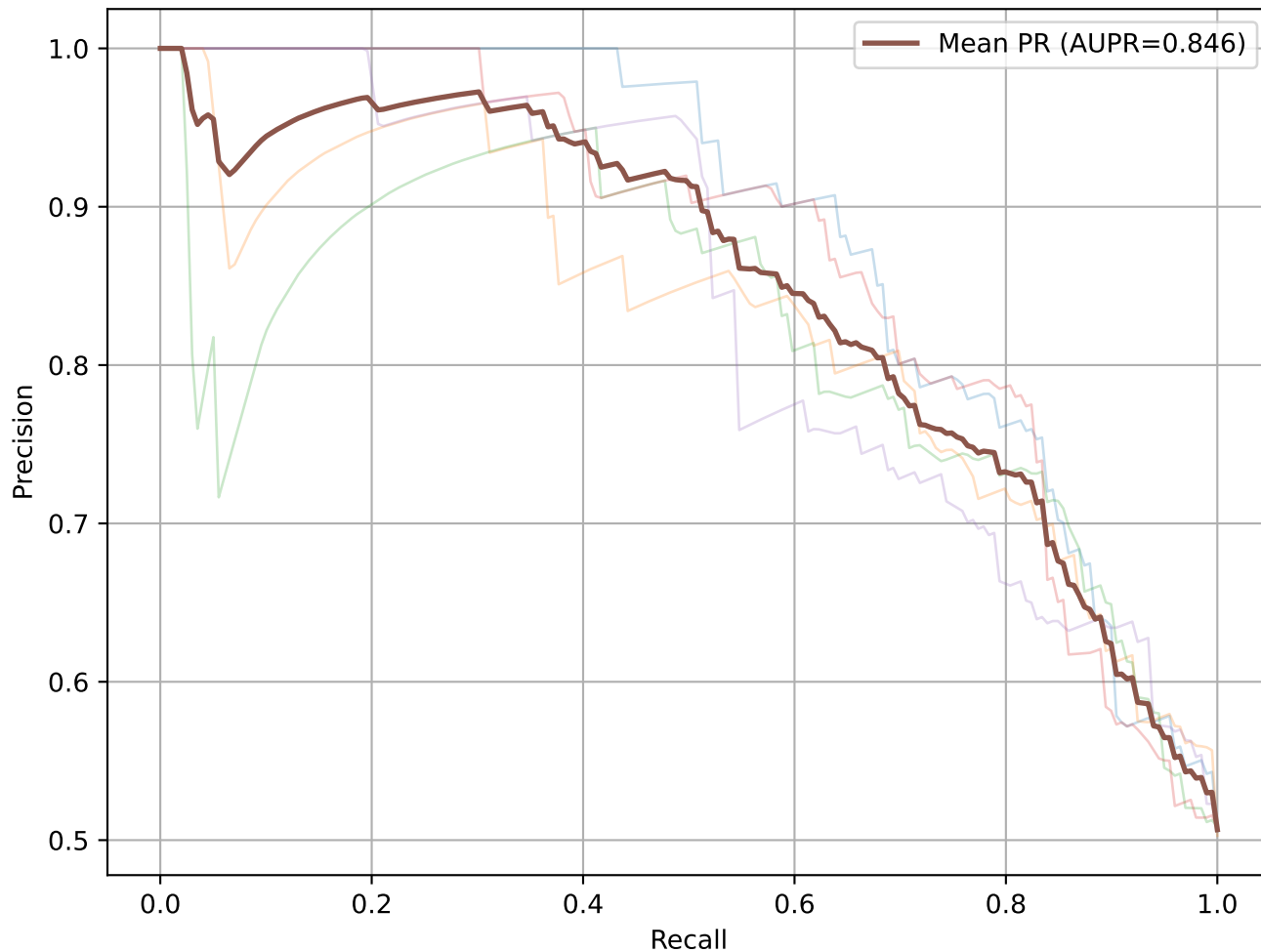
ClassificationReport - LOGISTIC (last fold)

	precision	recall	f1-score	support
0	0.79	0.80	0.79	89
1	0.80	0.79	0.80	92
accuracy			0.80	181
macro avg	0.80	0.80	0.80	181
weighted avg	0.80	0.80	0.80	181

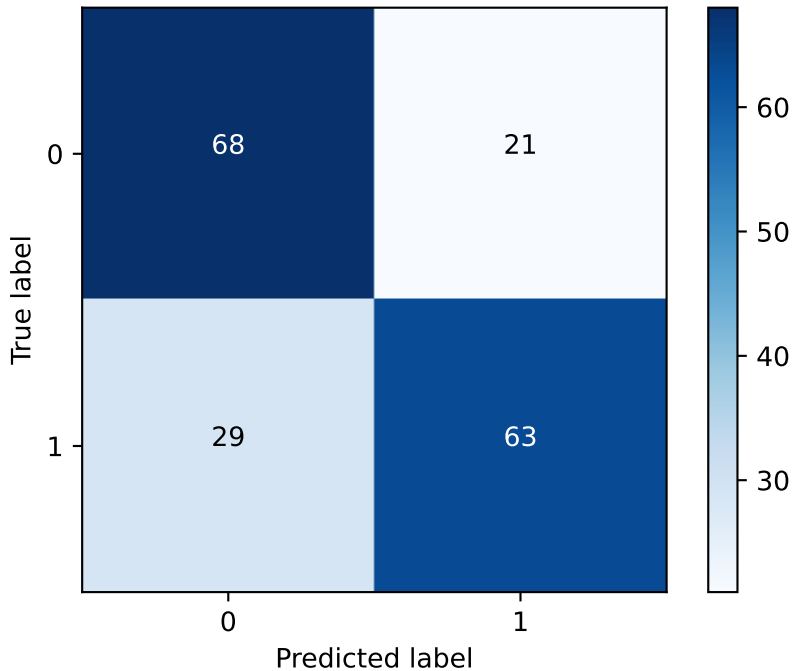
ROC - RF



Precision-Recall - RF



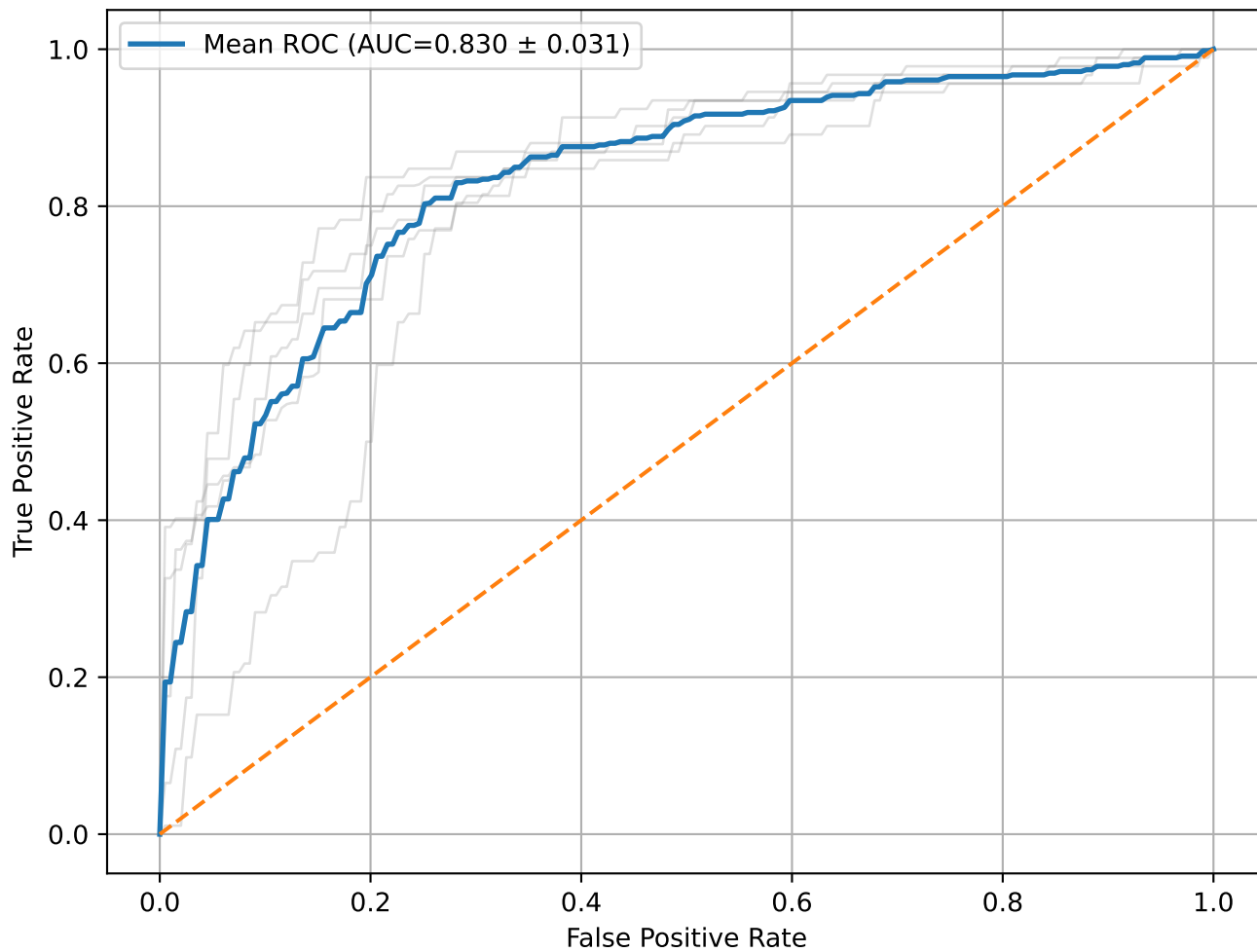
Confusion Matrix - RF (last fold)



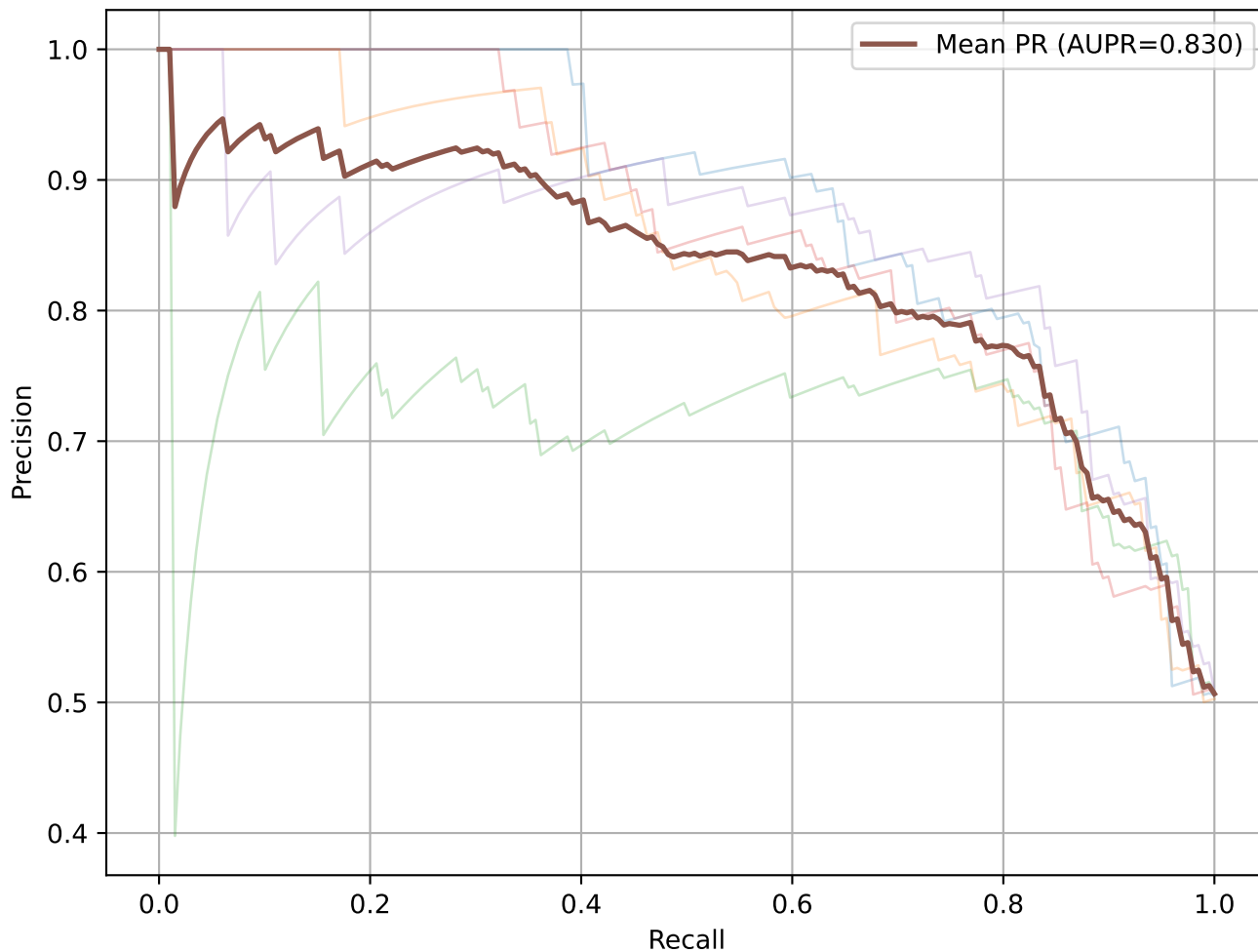
ClassificationReport - RF (last fold)

	precision	recall	f1-score	support
0	0.70	0.76	0.73	89
1	0.75	0.68	0.72	92
accuracy			0.72	181
macro avg	0.73	0.72	0.72	181
weighted avg	0.73	0.72	0.72	181

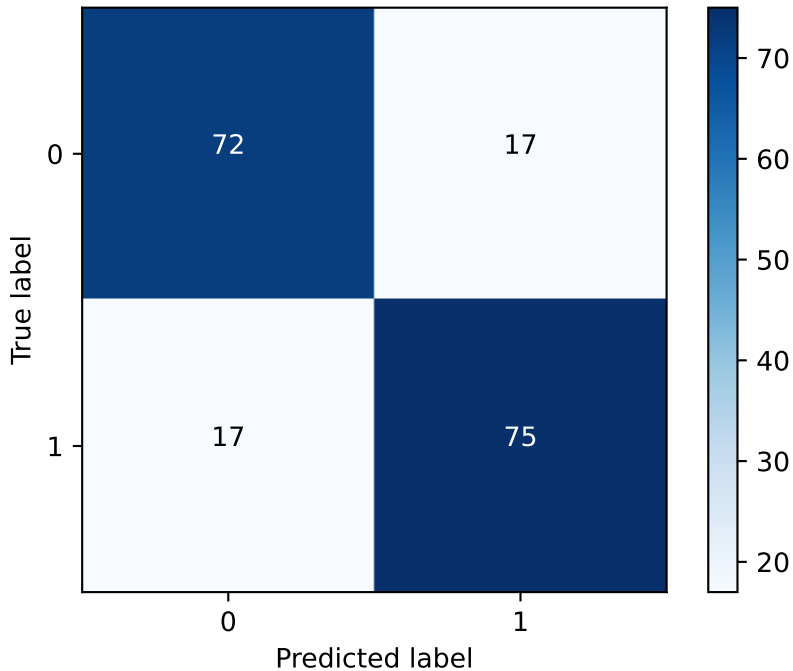
ROC - SVM



Precision-Recall - SVM



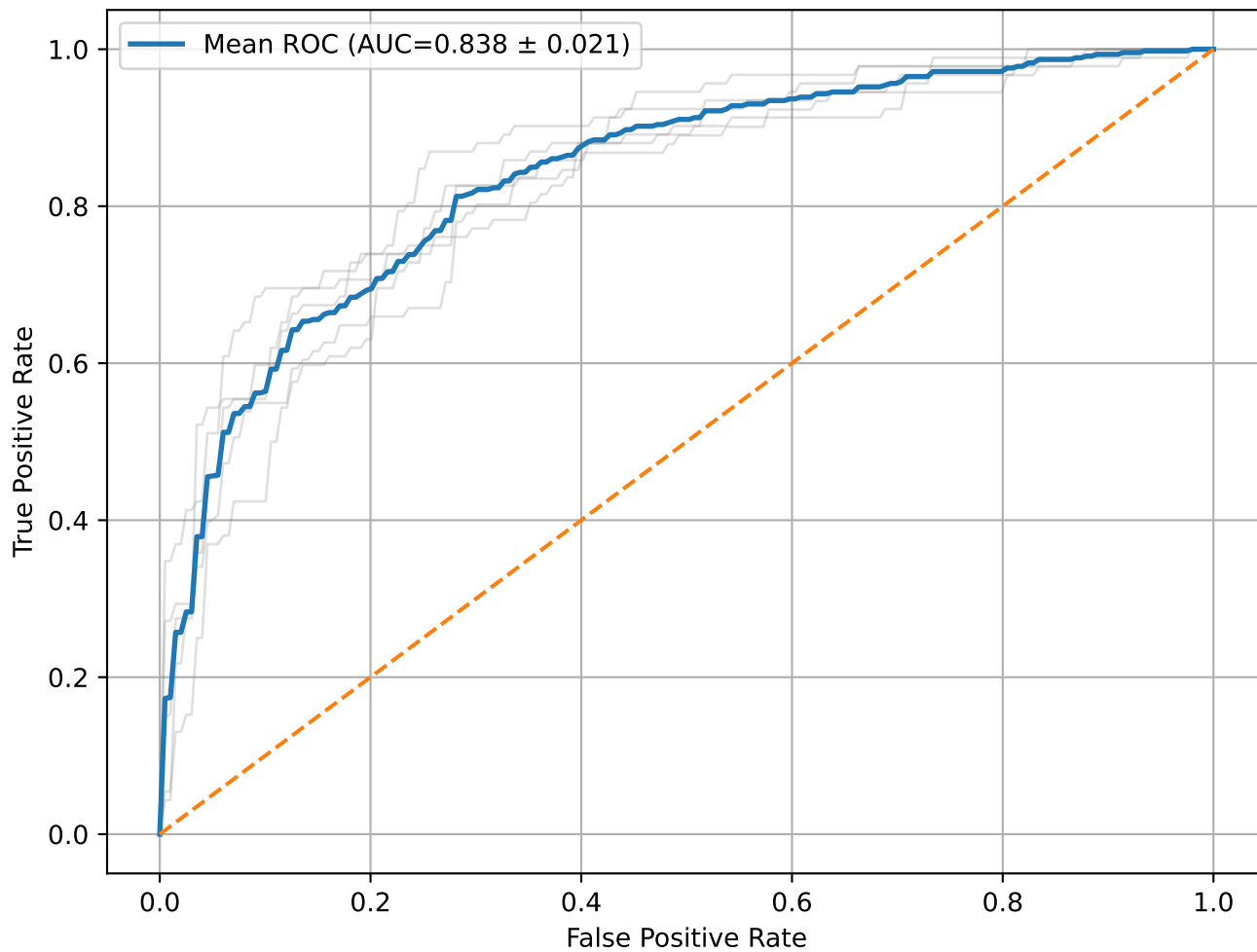
Confusion Matrix - SVM (last fold)



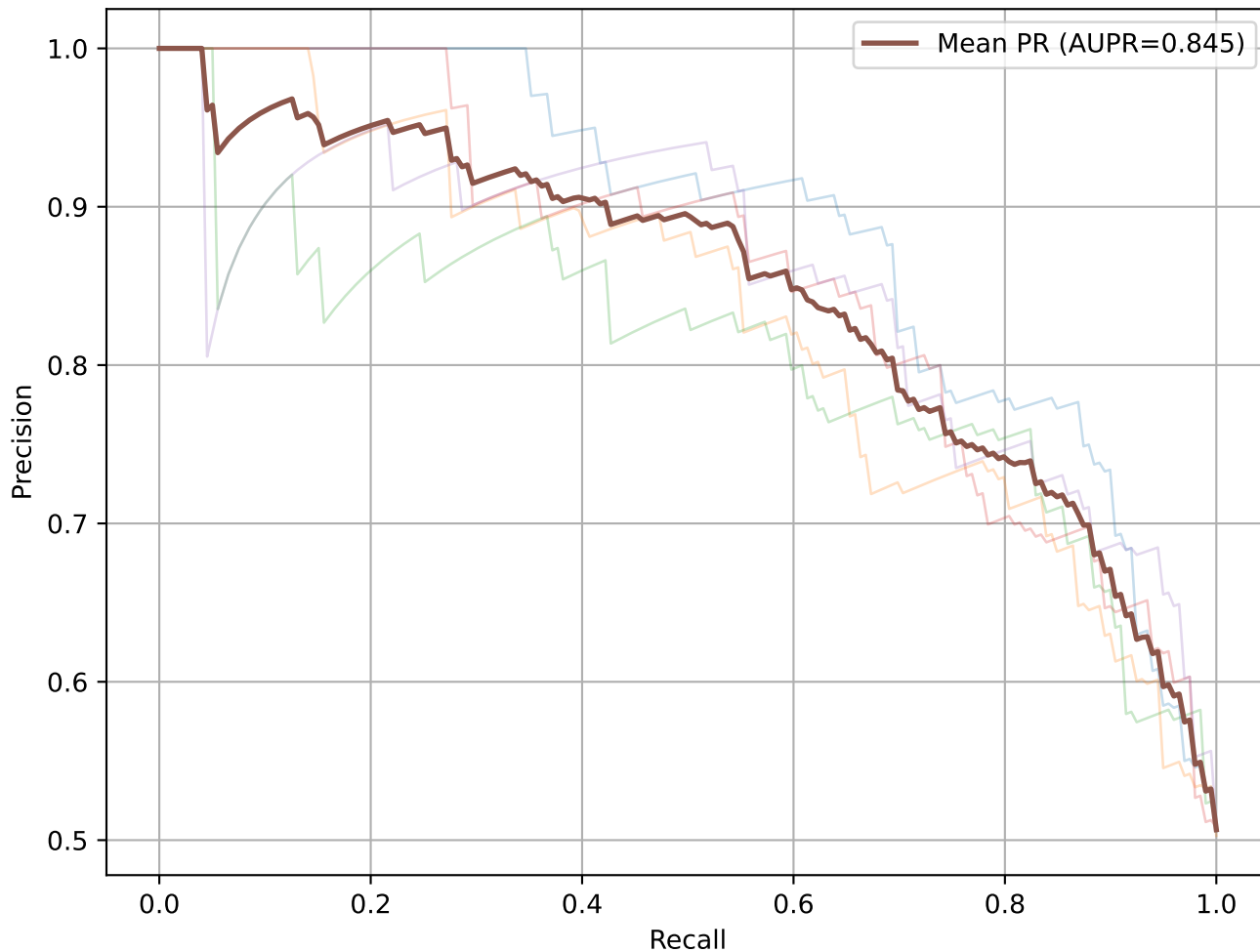
ClassificationReport - SVM (last fold)

	precision	recall	f1-score	support
0	0.81	0.81	0.81	89
1	0.82	0.82	0.82	92
accuracy			0.81	181
macro avg	0.81	0.81	0.81	181
weighted avg	0.81	0.81	0.81	181

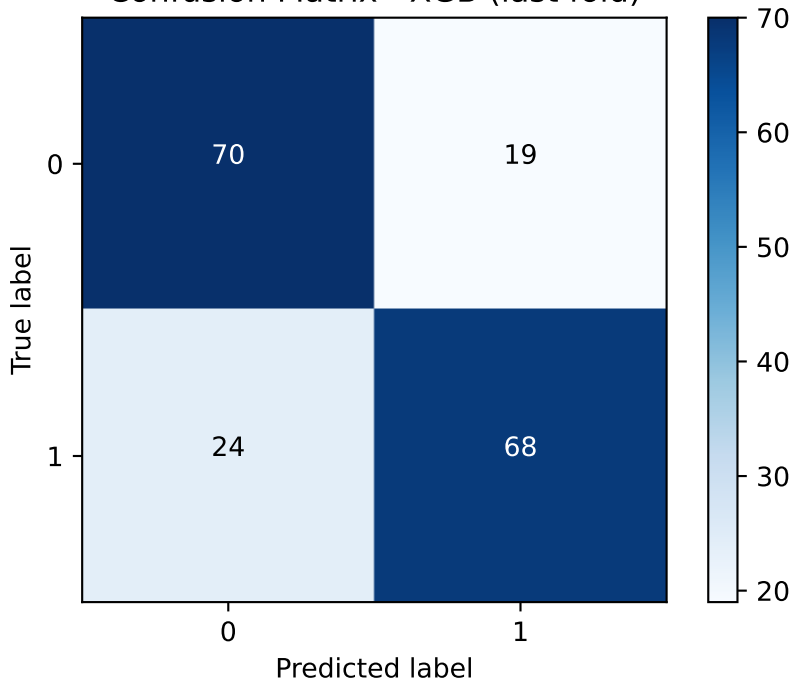
ROC - XGB



Precision-Recall - XGB



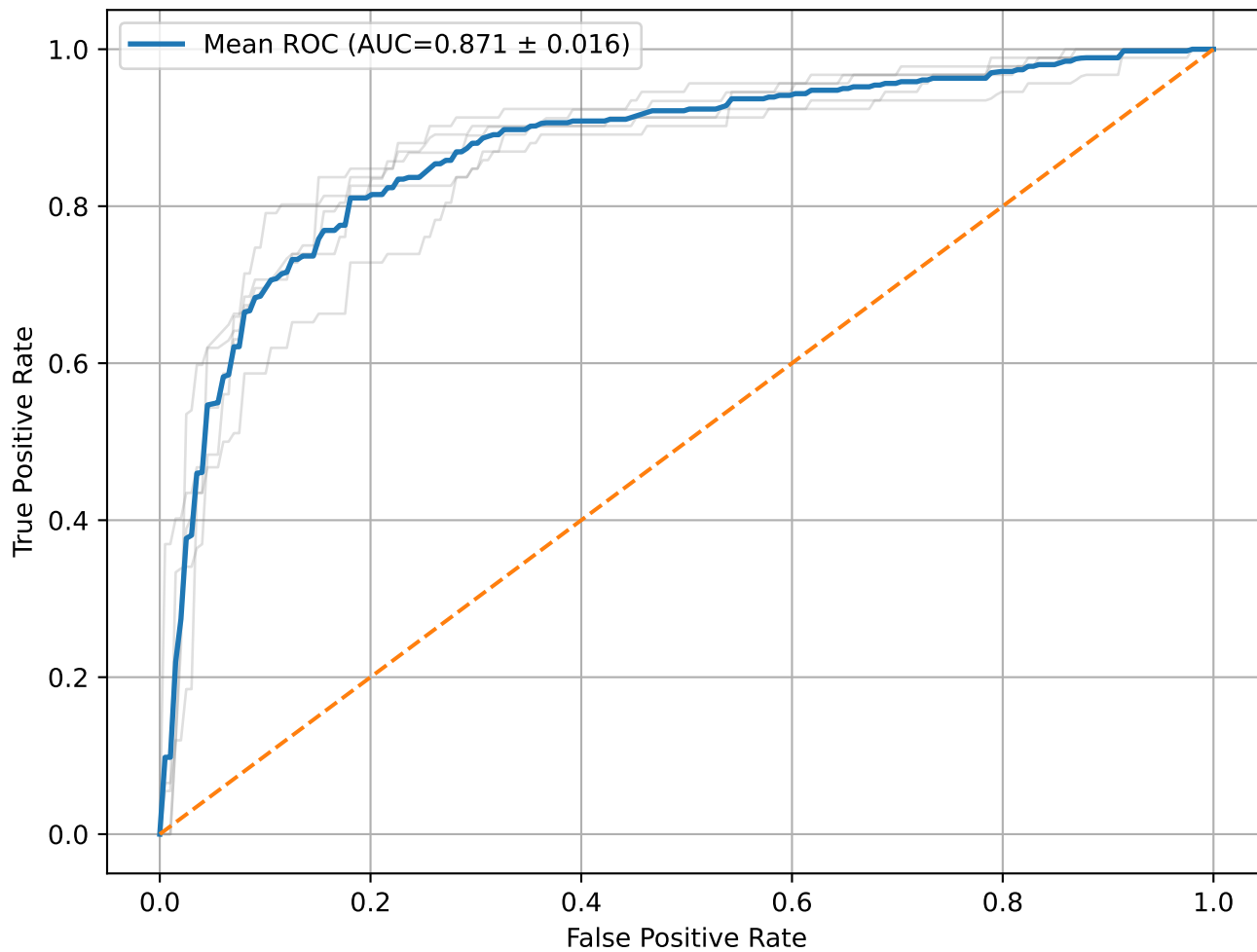
Confusion Matrix - XGB (last fold)



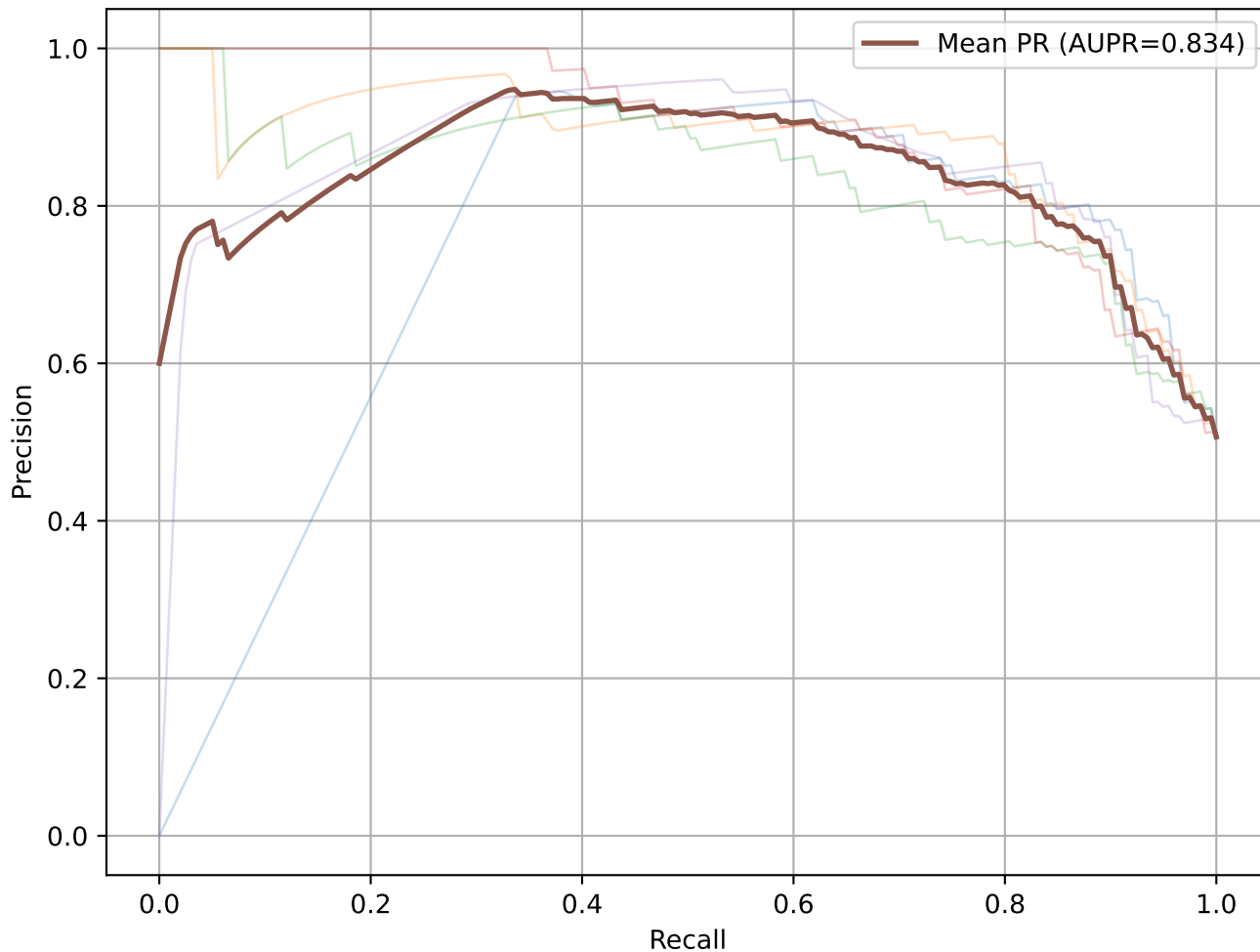
ClassificationReport - XGB (last fold)

	precision	recall	f1-score	support
0	0.74	0.79	0.77	89
1	0.78	0.74	0.76	92
accuracy			0.76	181
macro avg	0.76	0.76	0.76	181
weighted avg	0.76	0.76	0.76	181

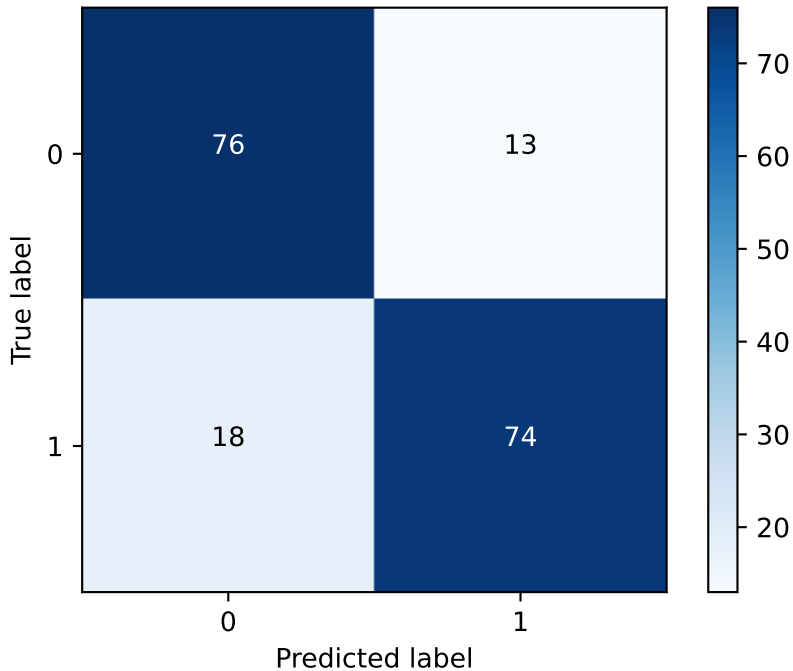
ROC - LGB



Precision-Recall - LGB



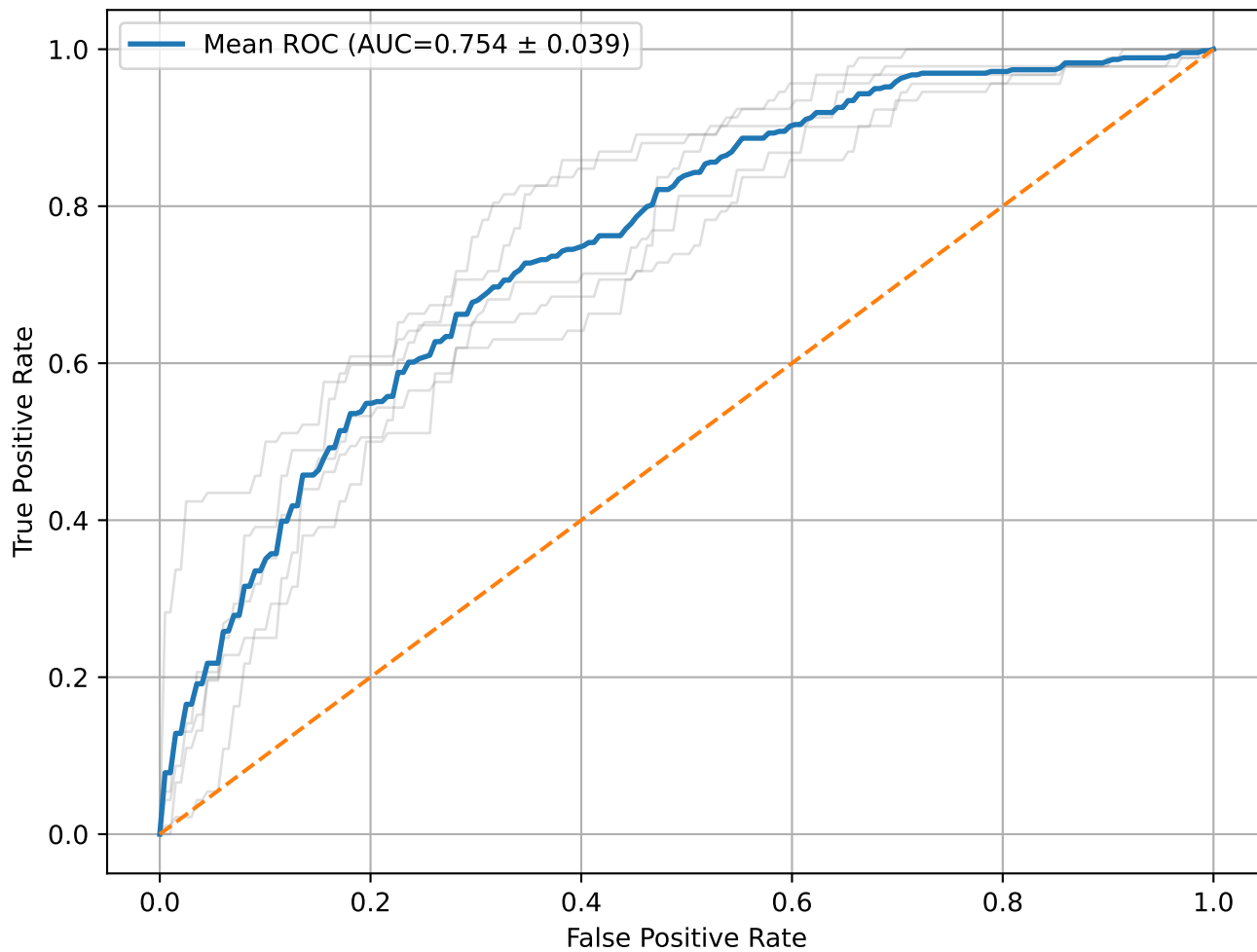
Confusion Matrix - LGB (last fold)



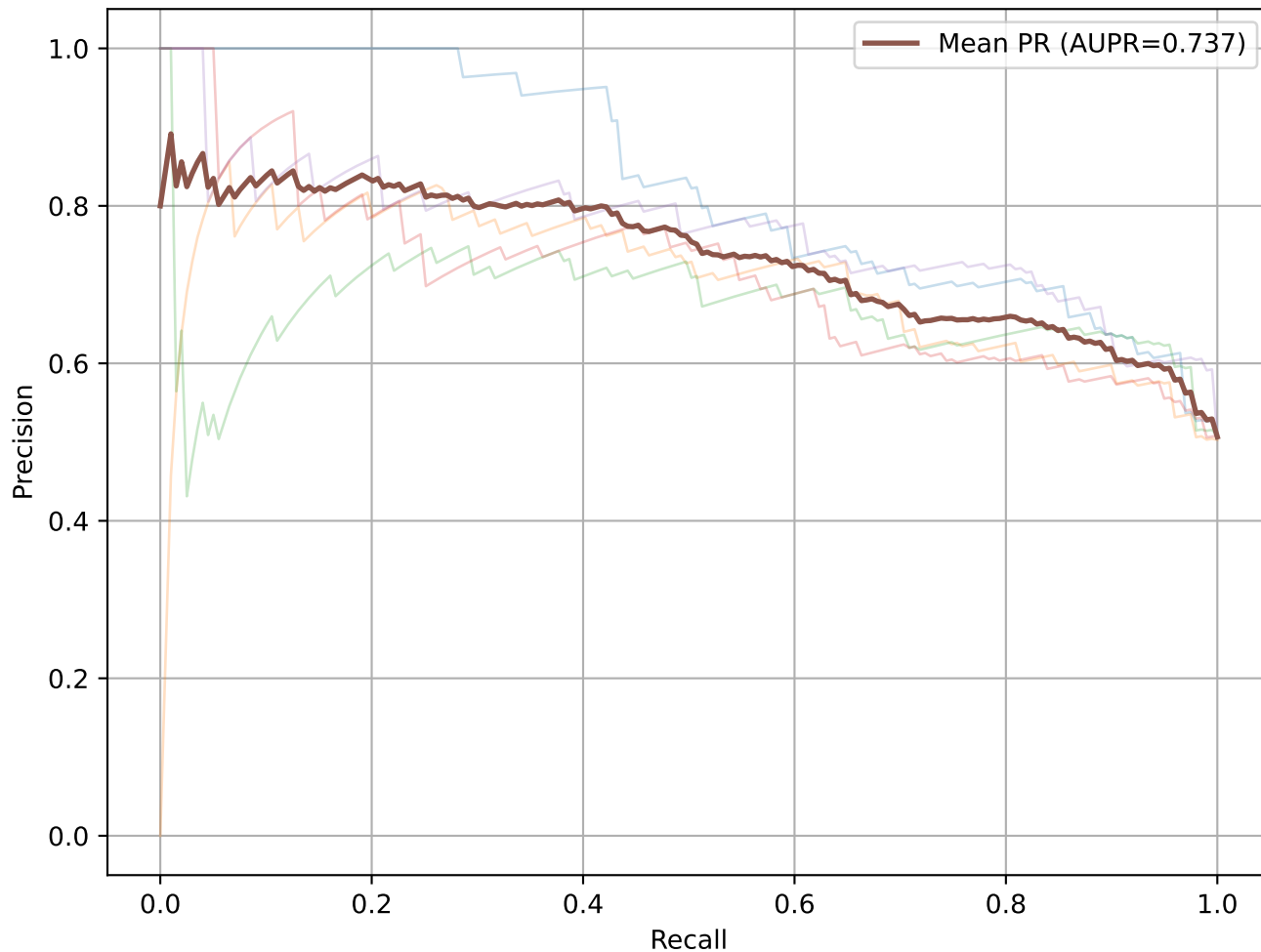
ClassificationReport - LGB (last fold)

	precision	recall	f1-score	support
0	0.81	0.85	0.83	89
1	0.85	0.80	0.83	92
accuracy			0.83	181
macro avg	0.83	0.83	0.83	181
weighted avg	0.83	0.83	0.83	181

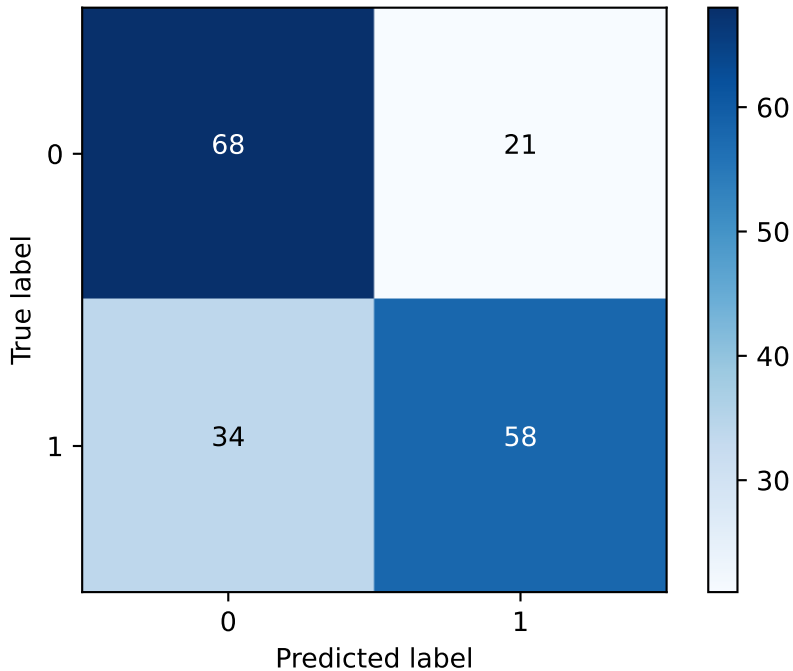
ROC - DNN



Precision-Recall - DNN



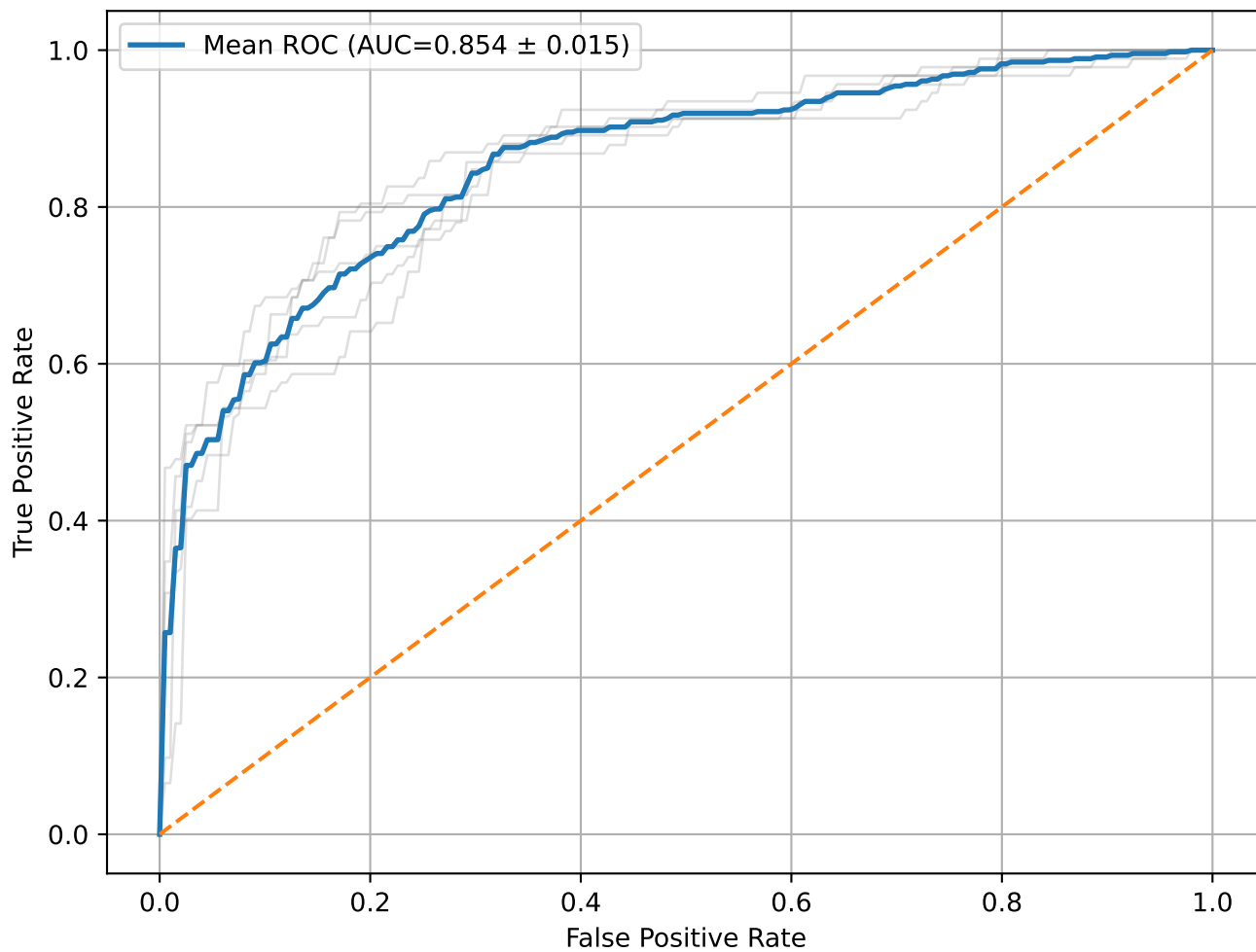
Confusion Matrix - DNN (last fold)



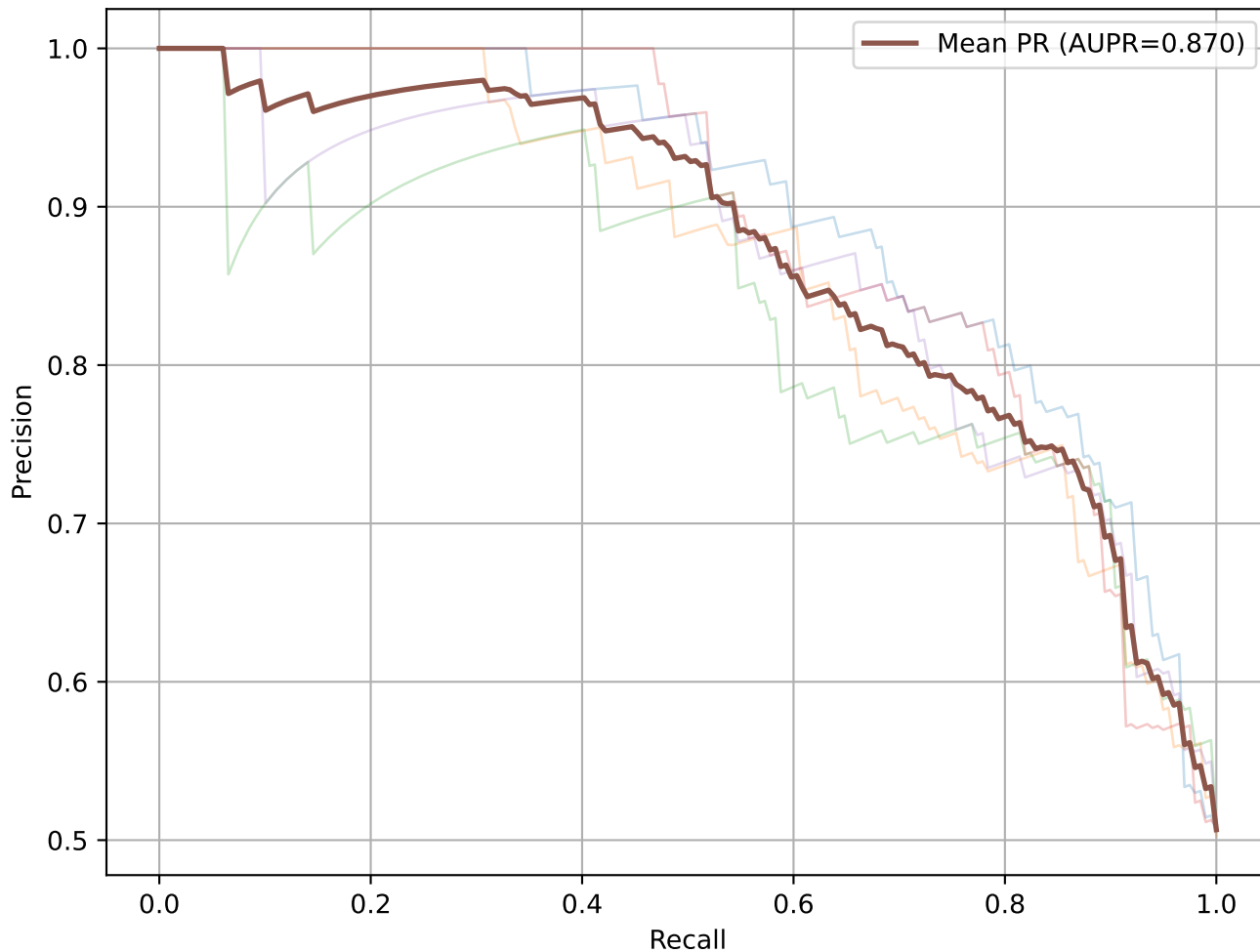
ClassificationReport - DNN (last fold)

	precision	recall	f1-score	support
0	0.67	0.76	0.71	89
1	0.73	0.63	0.68	92
accuracy			0.70	181
macro avg	0.70	0.70	0.70	181
weighted avg	0.70	0.70	0.69	181

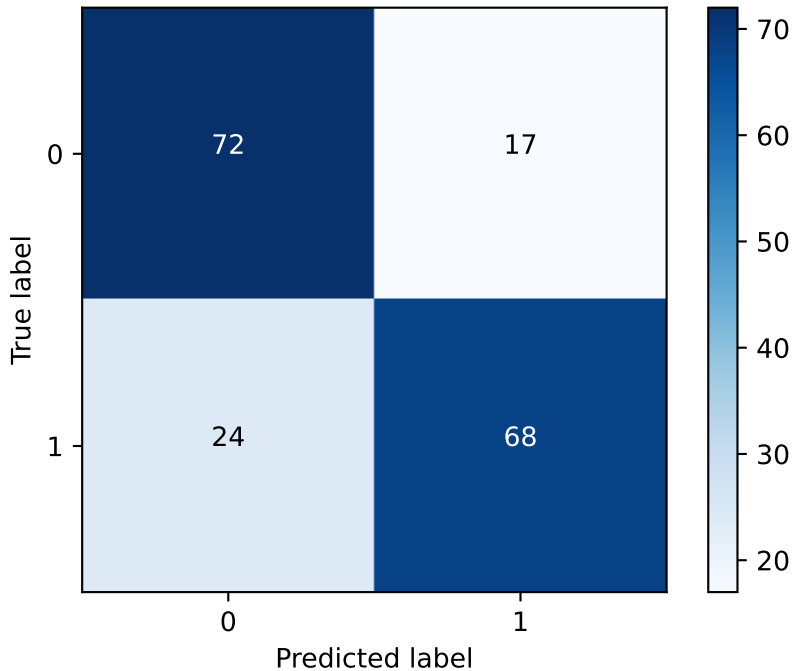
ROC - STACKING



Precision-Recall - STACKING



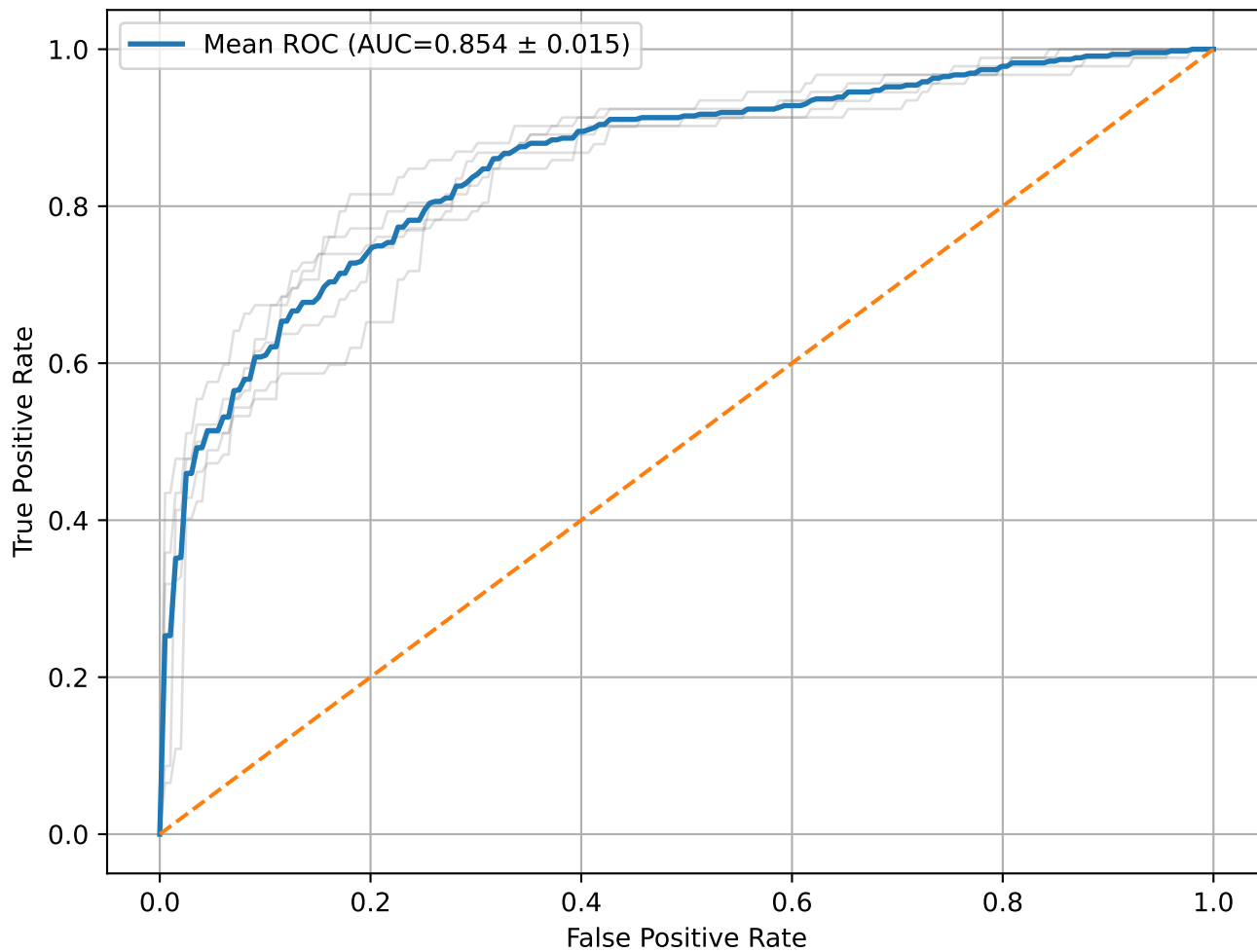
Confusion Matrix - STACKING (last fold)



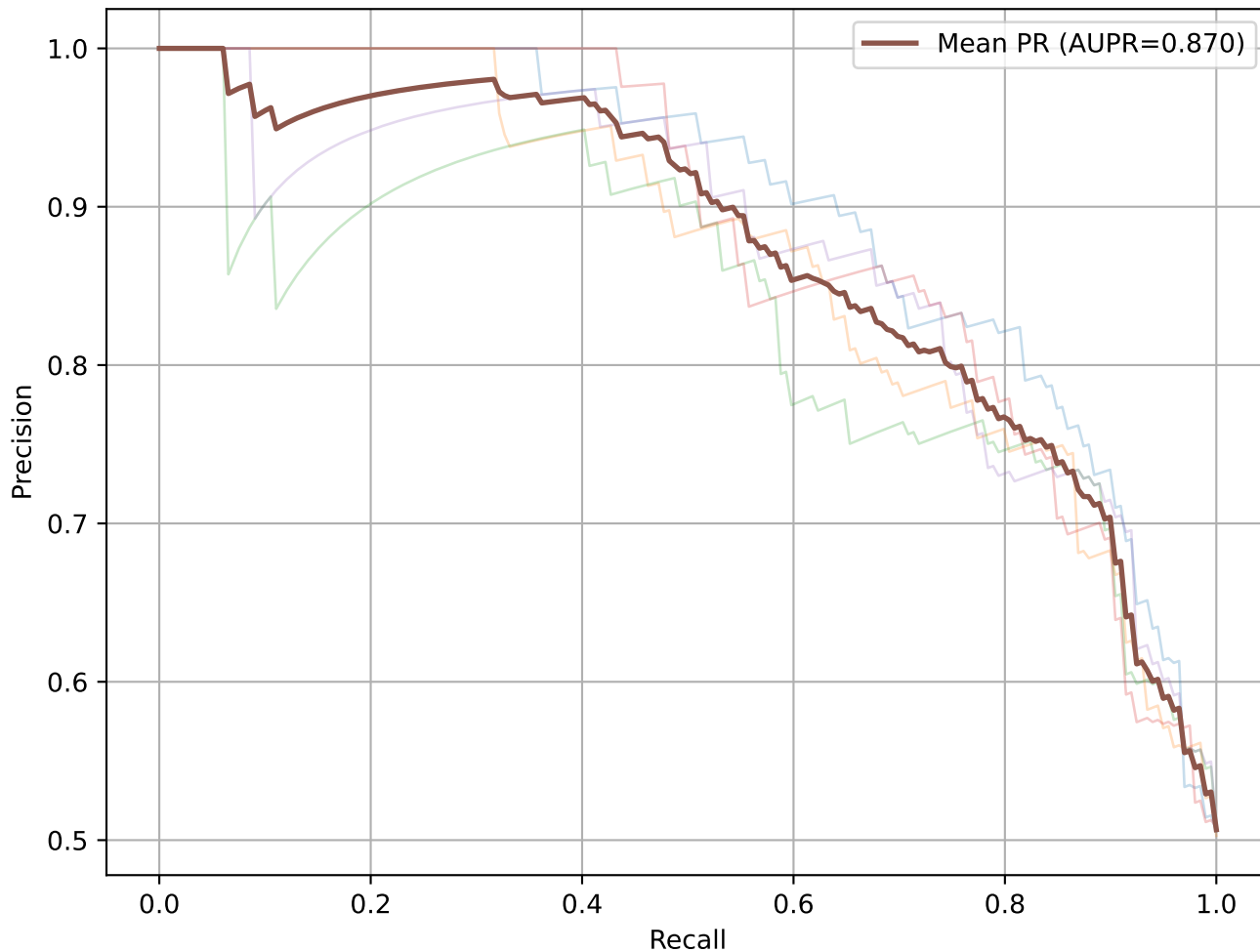
ClassificationReport - STACKING (last fold)

	precision	recall	f1-score	support
0	0.75	0.81	0.78	89
1	0.80	0.74	0.77	92
accuracy			0.77	181
macro avg	0.78	0.77	0.77	181
weighted avg	0.78	0.77	0.77	181

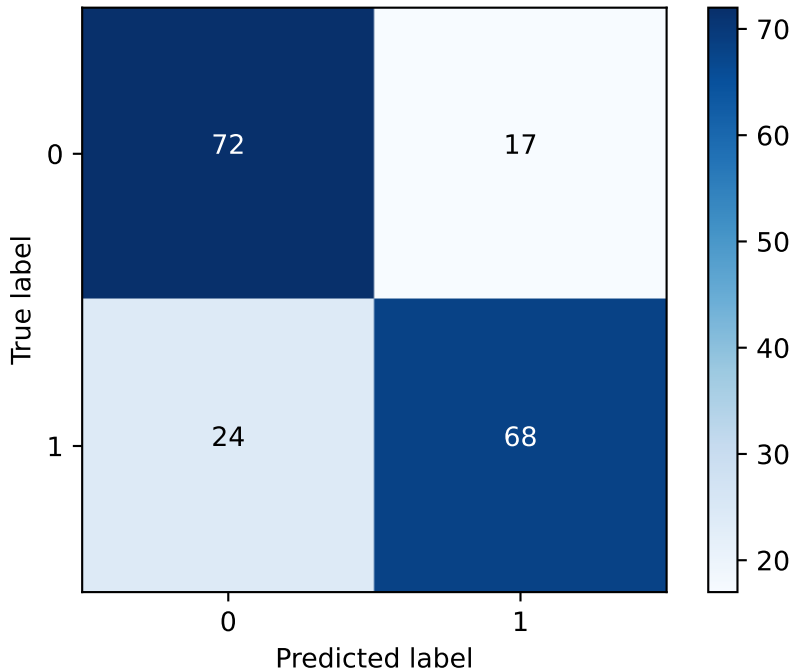
ROC - VOTING



ROC curve for Mean PR (AUPR=0.870). The x-axis represents the False Positive Rate (FPR) and the y-axis represents the True Positive Rate (TPR), both ranging from 0.0 to 1.0. The curve shows a high performance, with the AUPR value of 0.870. The curve is a thick brown line, and the area under the curve is shaded in light brown. The legend indicates 'Mean PR (AUPR=0.870)'.



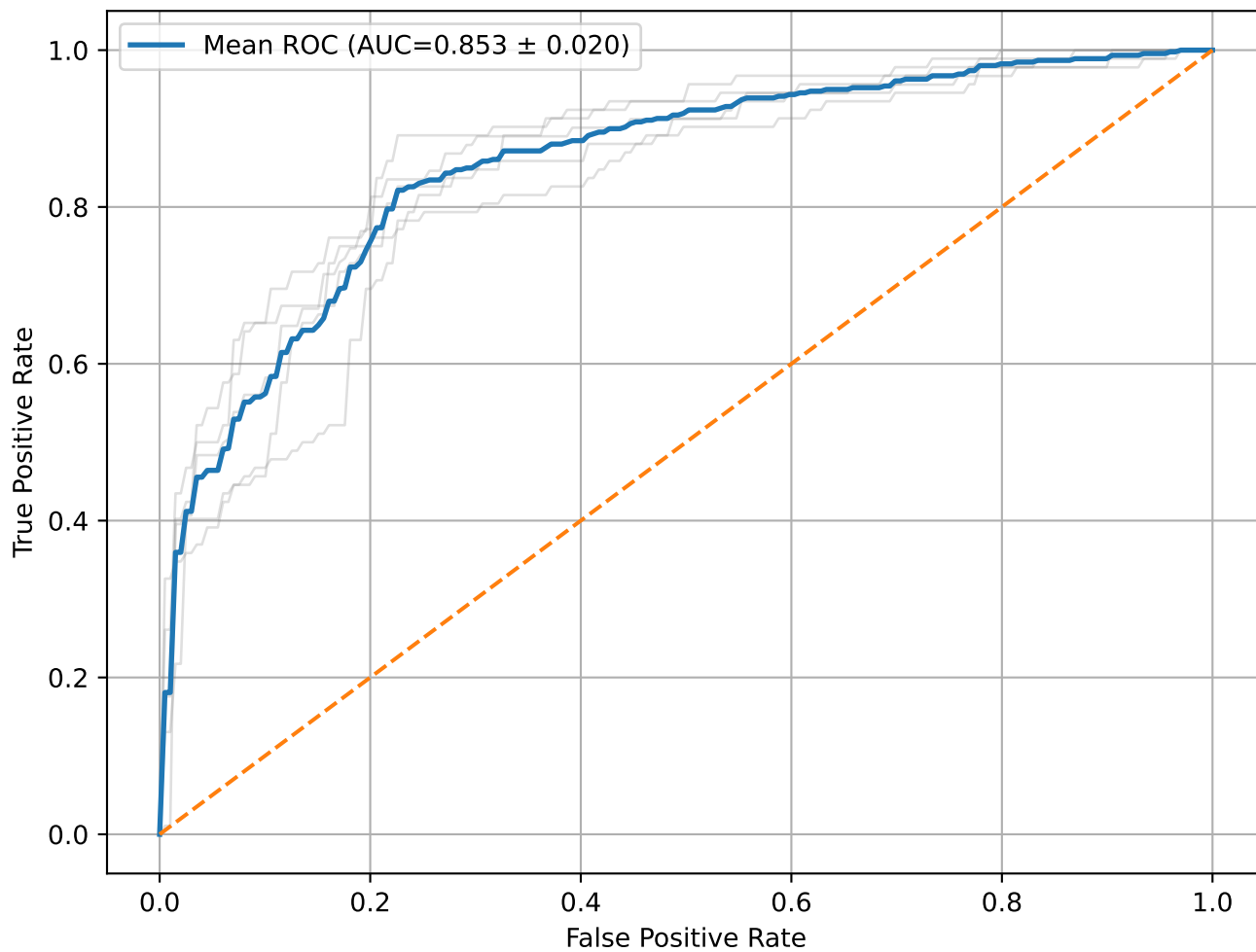
Confusion Matrix - VOTING (last fold)



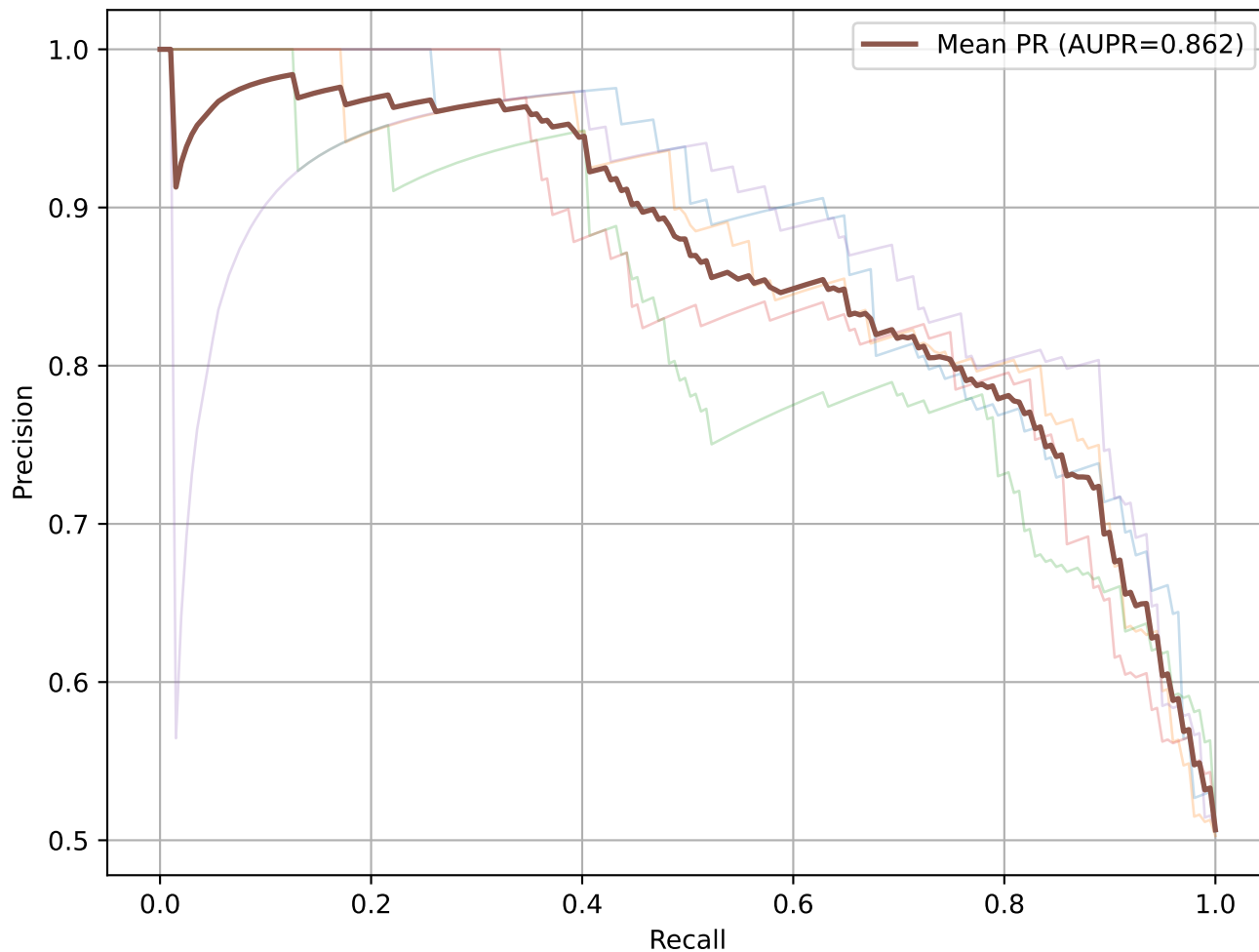
ClassificationReport - VOTING (last fold)

	precision	recall	f1-score	support
0	0.75	0.81	0.78	89
1	0.80	0.74	0.77	92
accuracy			0.77	181
macro avg	0.78	0.77	0.77	181
weighted avg	0.78	0.77	0.77	181

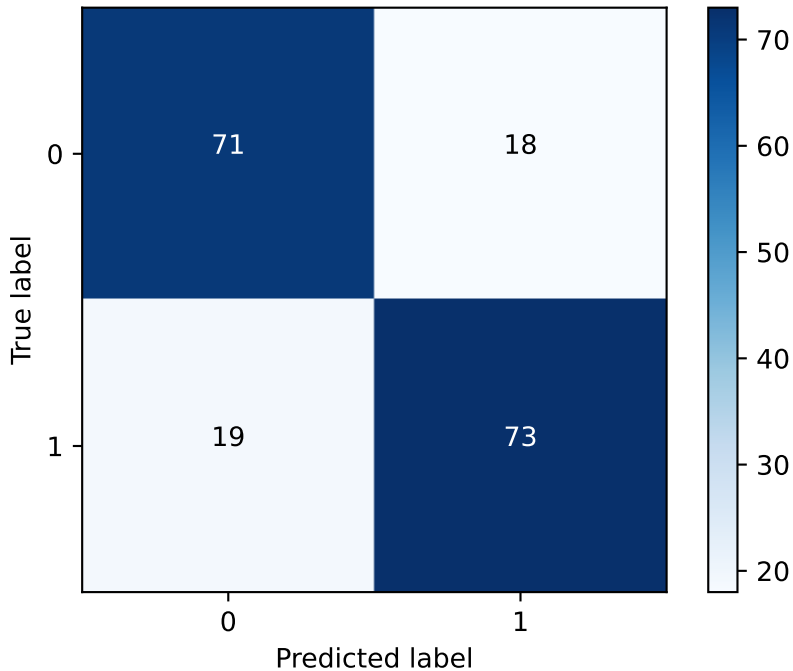
ROC - BAGGING



Precision-Recall - BAGGING



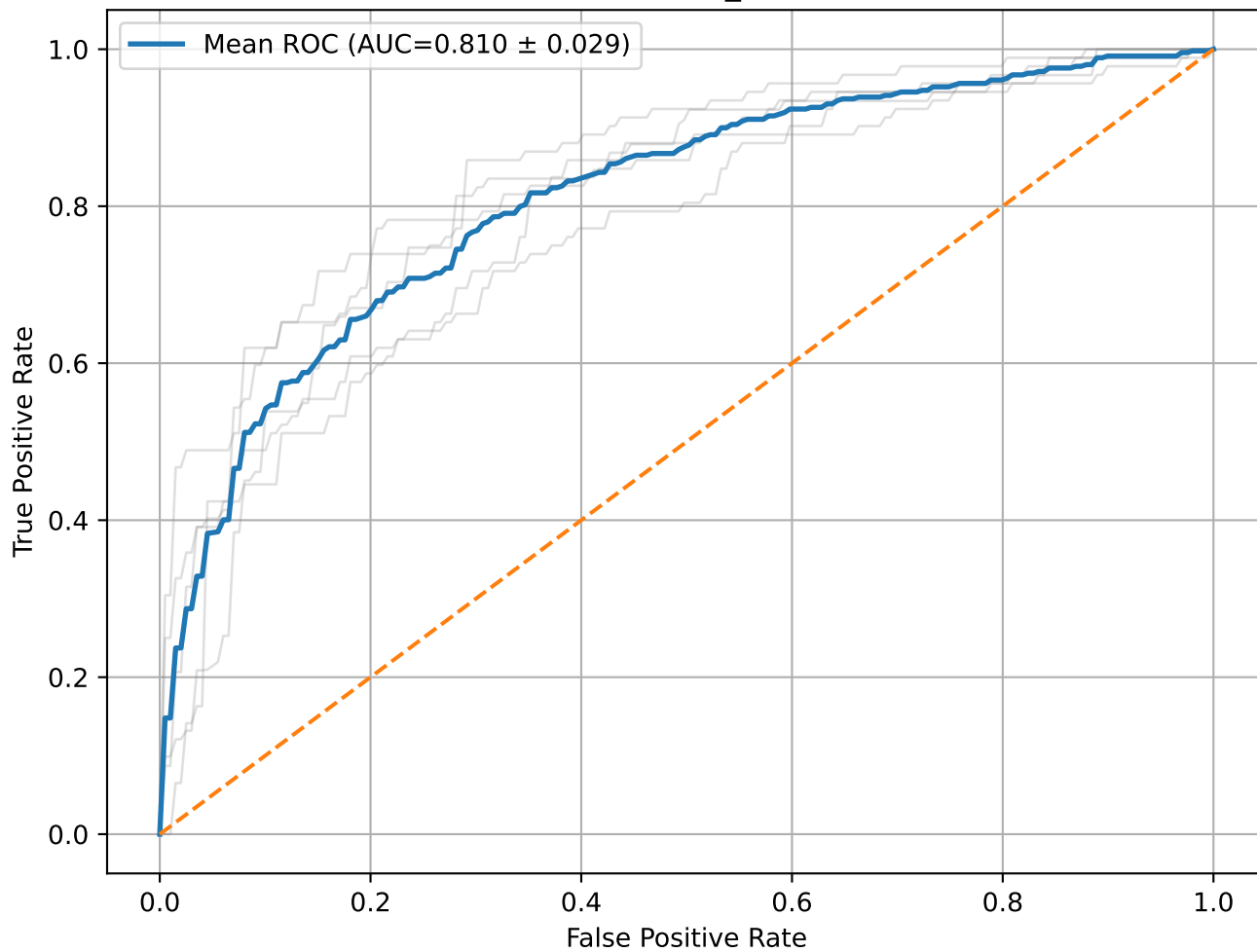
Confusion Matrix - BAGGING (last fold)



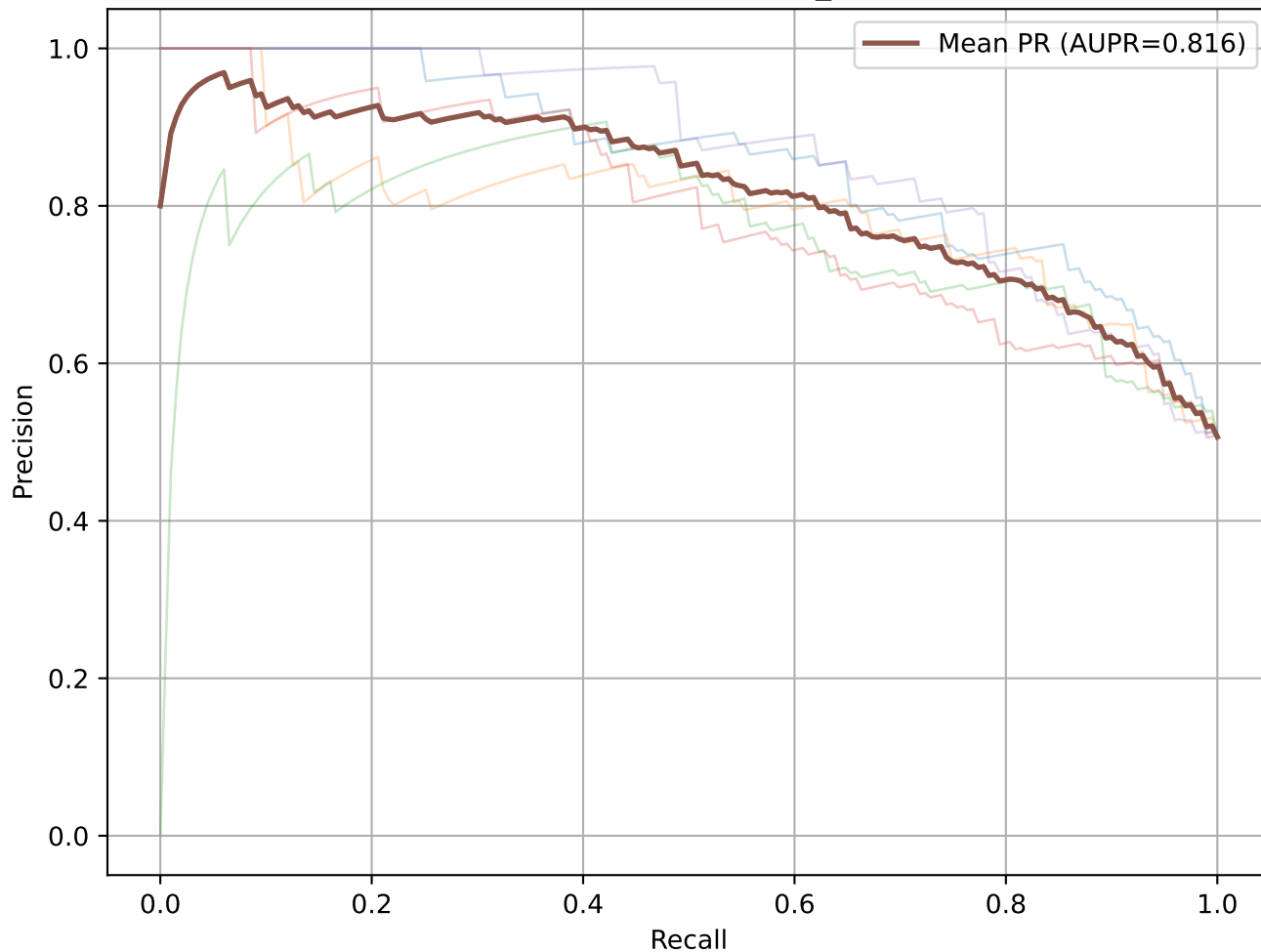
ClassificationReport - BAGGING (last fold)

	precision	recall	f1-score	support
0	0.79	0.80	0.79	89
1	0.80	0.79	0.80	92
accuracy			0.80	181
macro avg	0.80	0.80	0.80	181
weighted avg	0.80	0.80	0.80	181

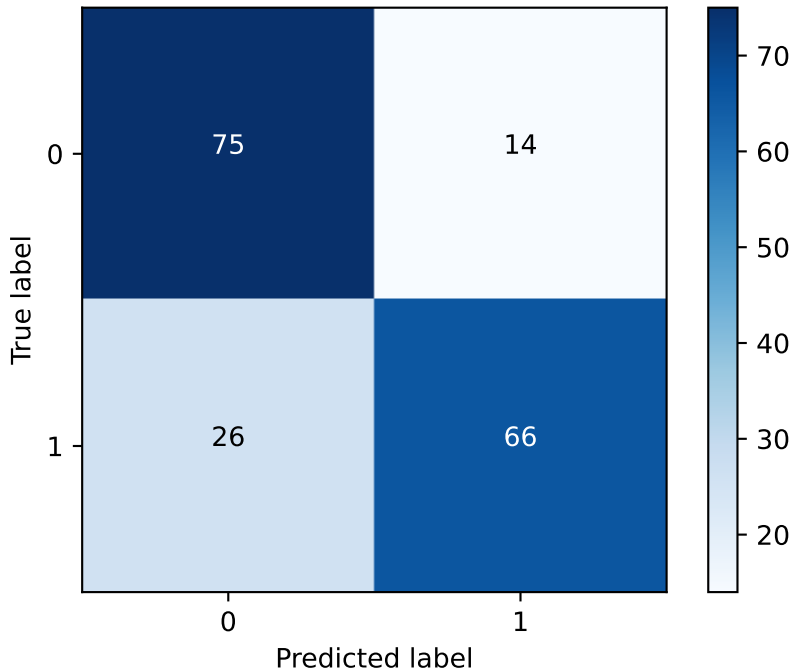
ROC - PCA_SVM



Precision-Recall - PCA_SVM



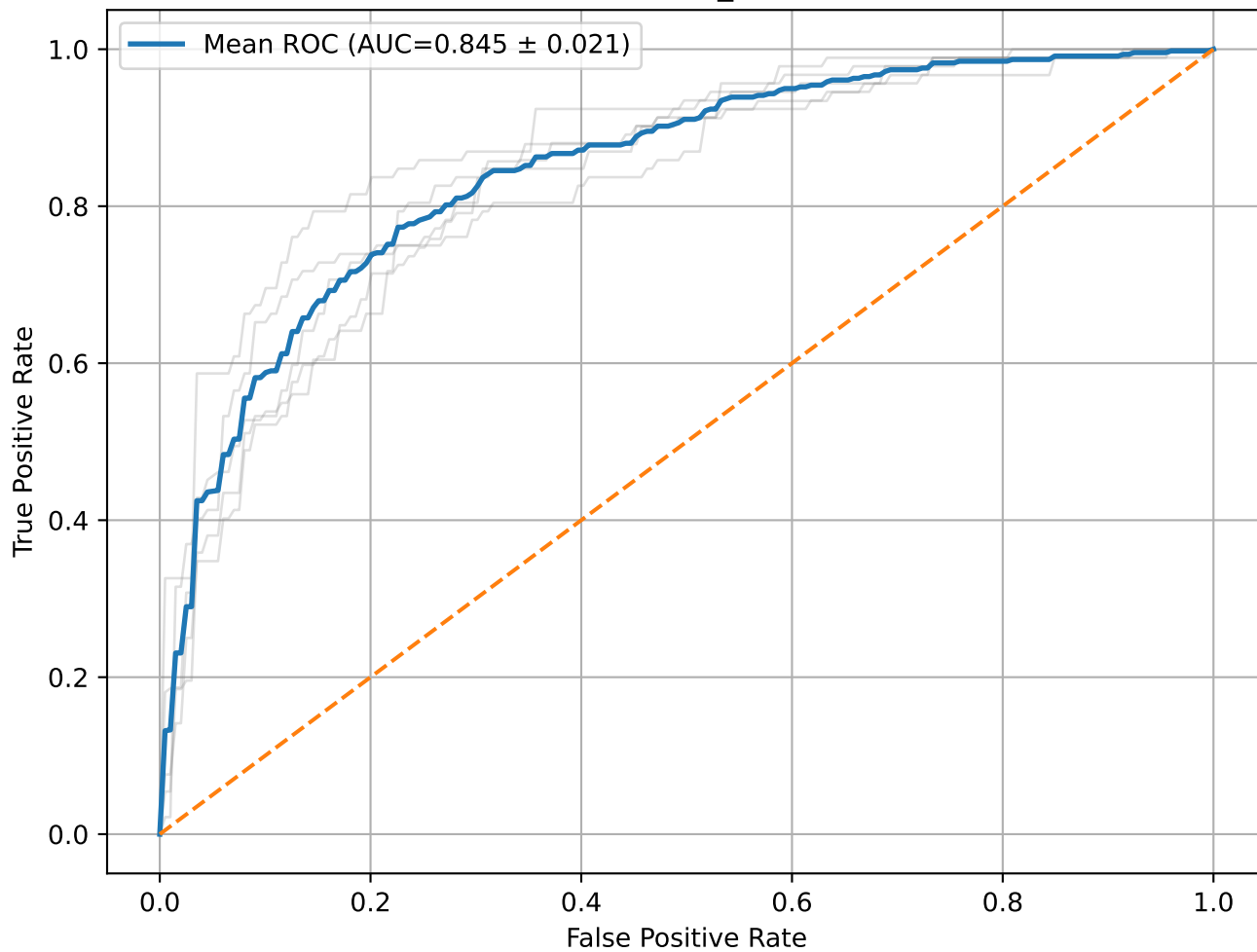
Confusion Matrix - PCA_SVM (last fold)



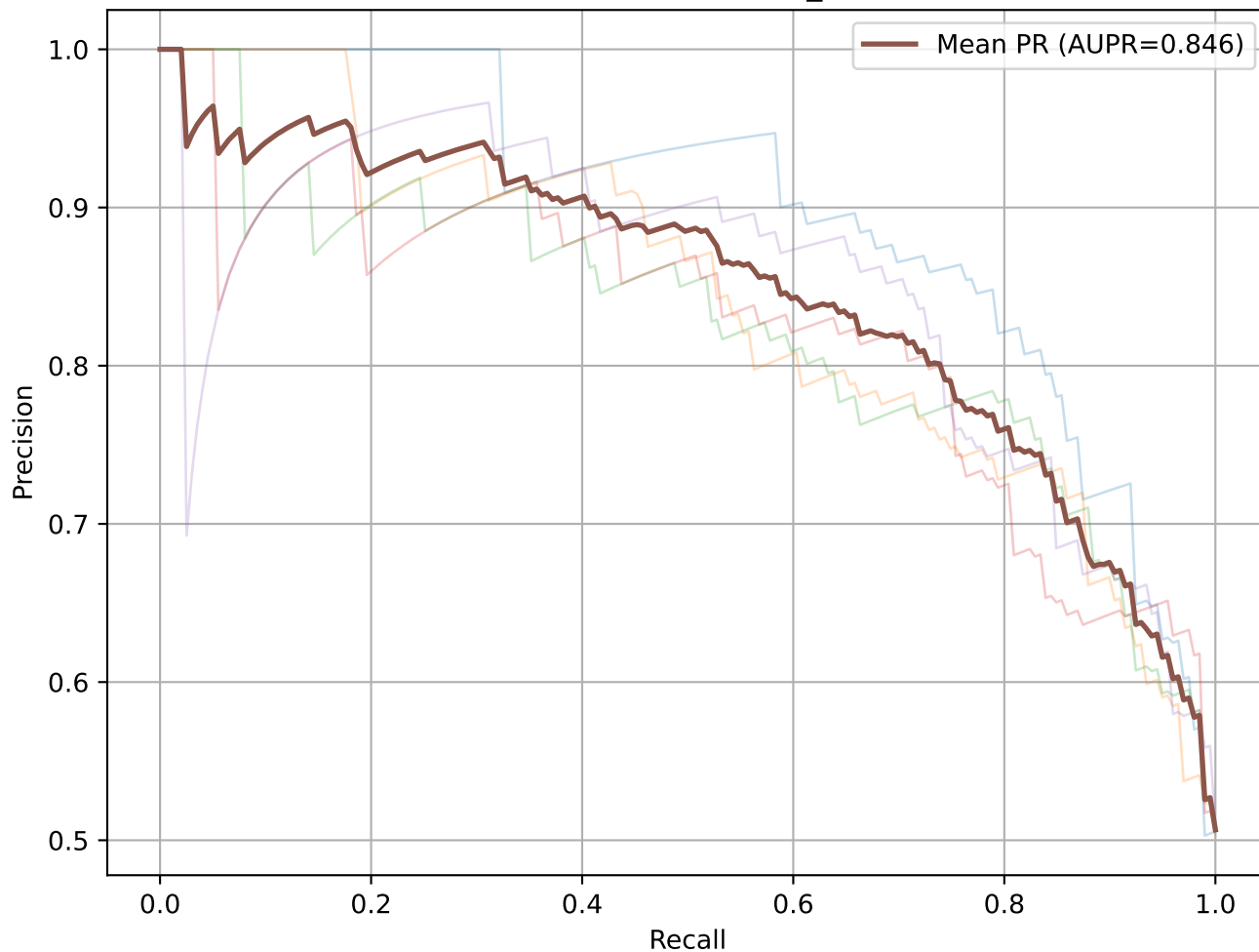
ClassificationReport - PCA_SVM (last fold)

	precision	recall	f1-score	support
0	0.74	0.84	0.79	89
1	0.82	0.72	0.77	92
accuracy			0.78	181
macro avg	0.78	0.78	0.78	181
weighted avg	0.78	0.78	0.78	181

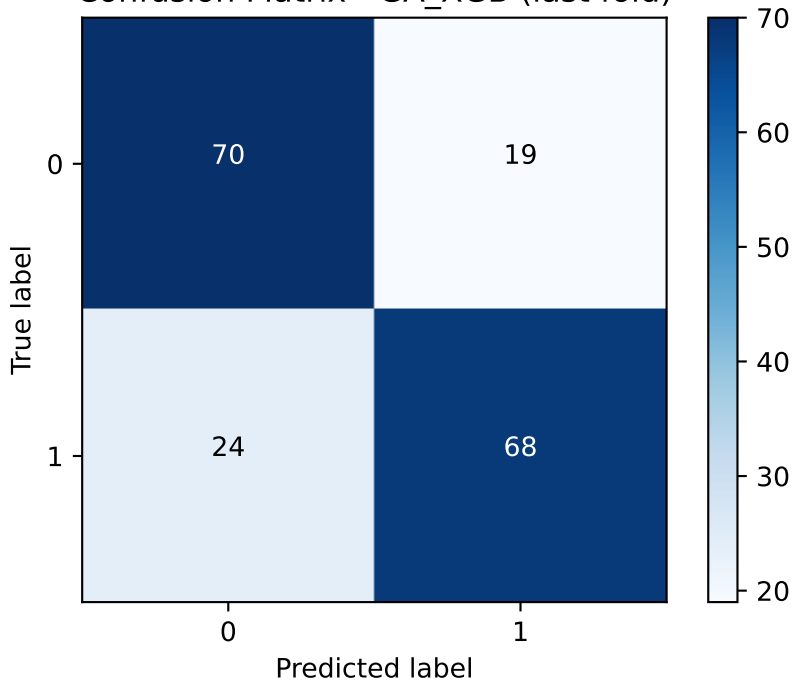
ROC - GA_XGB



Precision-Recall - GA_XGB



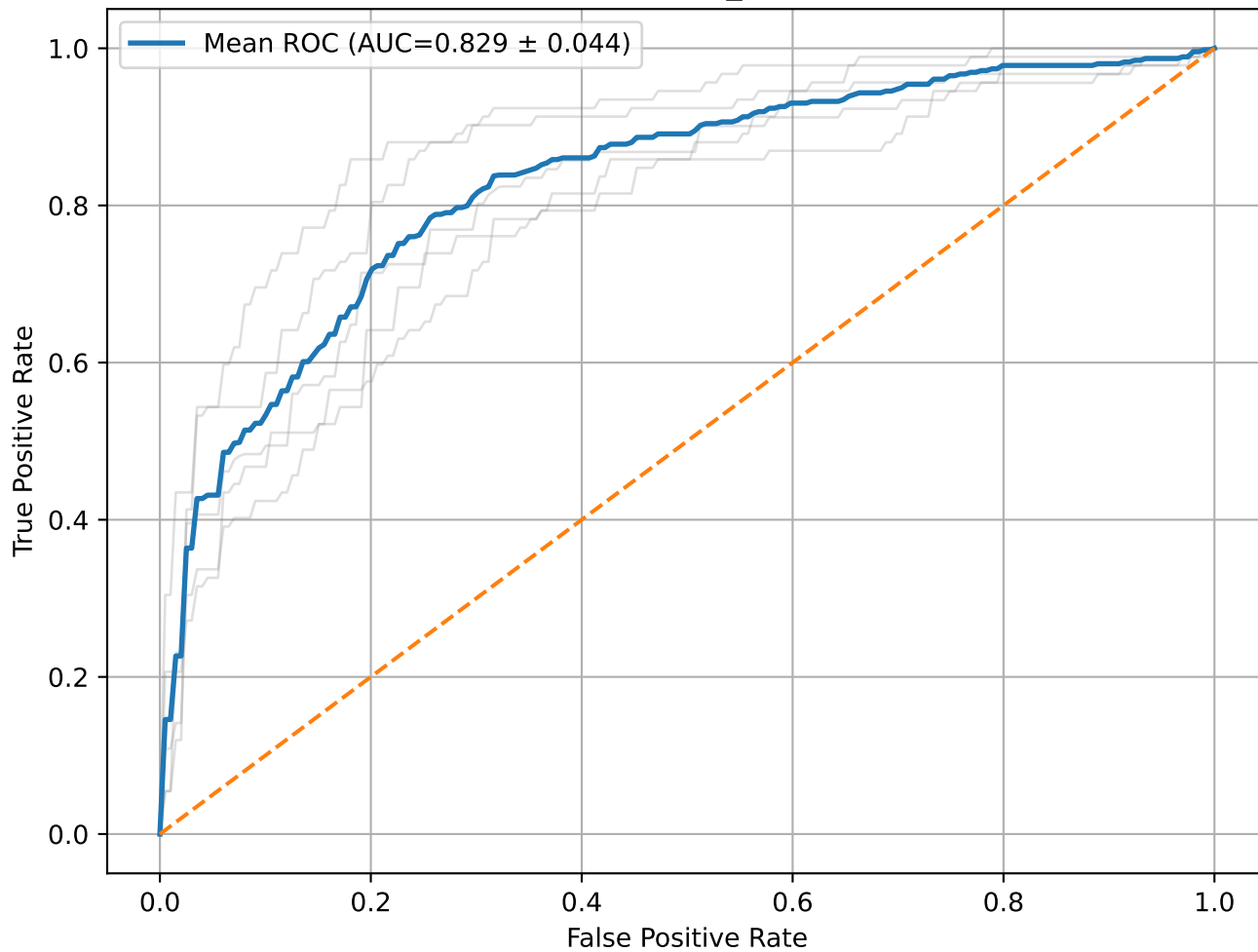
Confusion Matrix - GA_XGB (last fold)

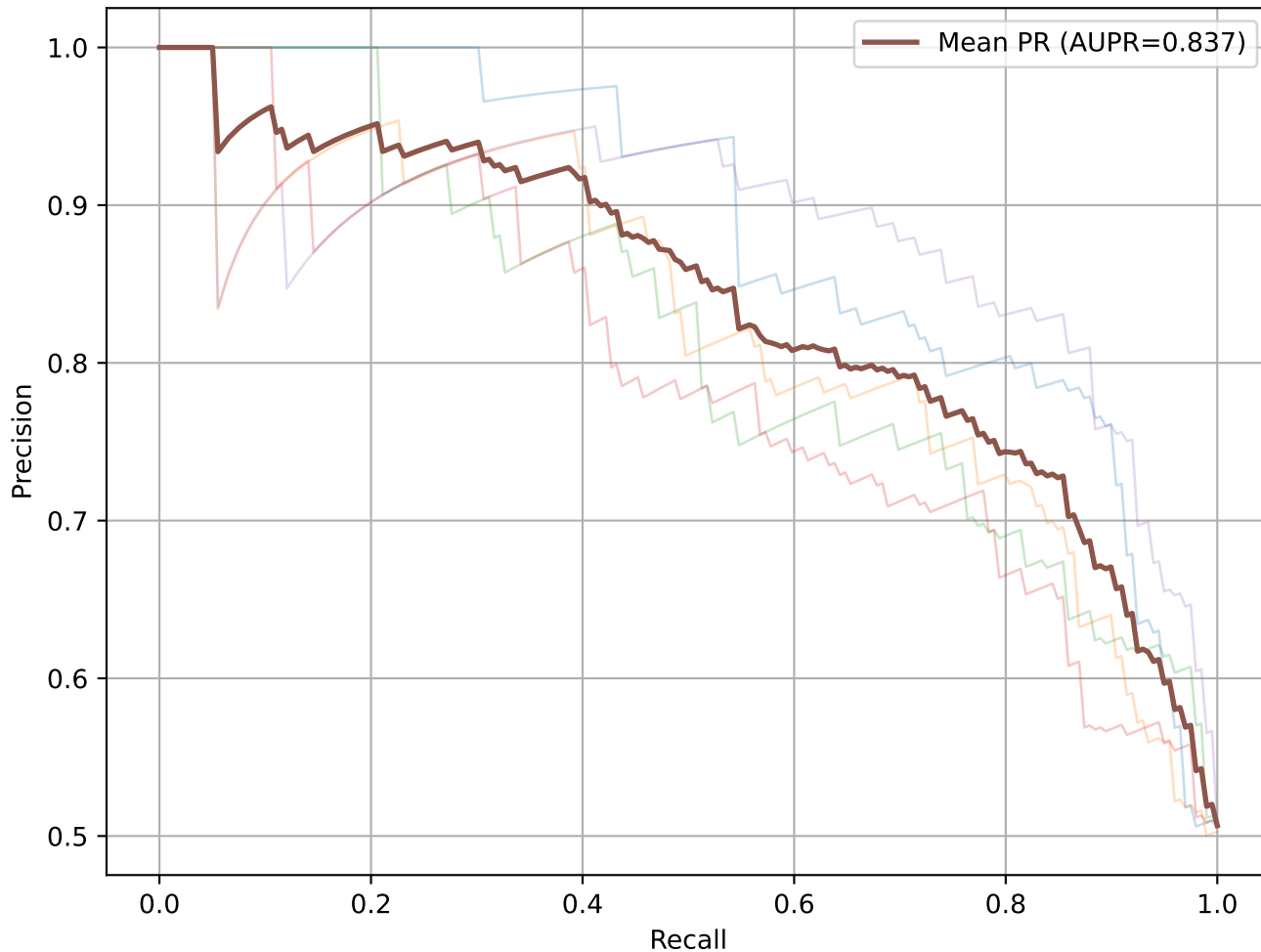


ClassificationReport - GA_XGB (last fold)

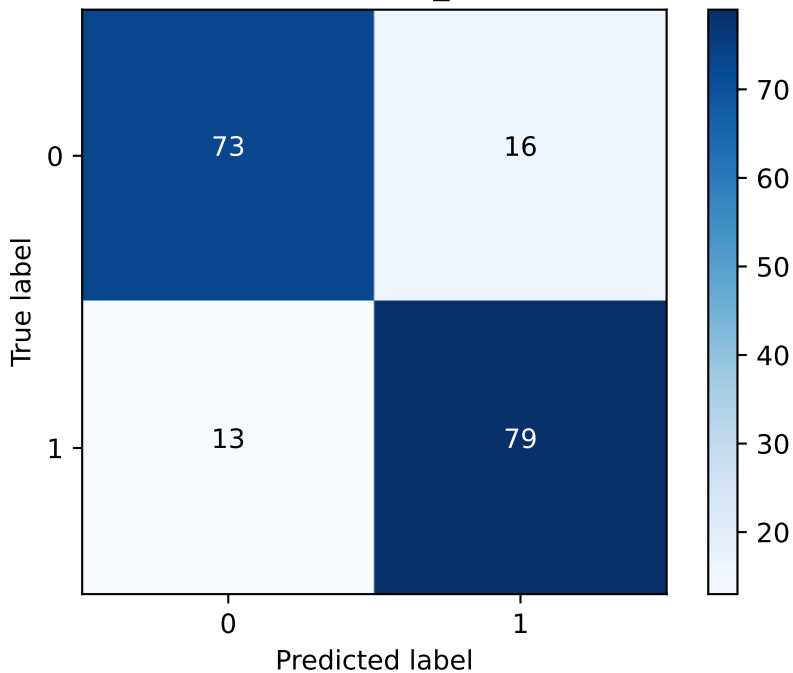
	precision	recall	f1-score	support
0	0.74	0.79	0.77	89
1	0.78	0.74	0.76	92
accuracy			0.76	181
macro avg	0.76	0.76	0.76	181
weighted avg	0.76	0.76	0.76	181

ROC - PSO_SVM





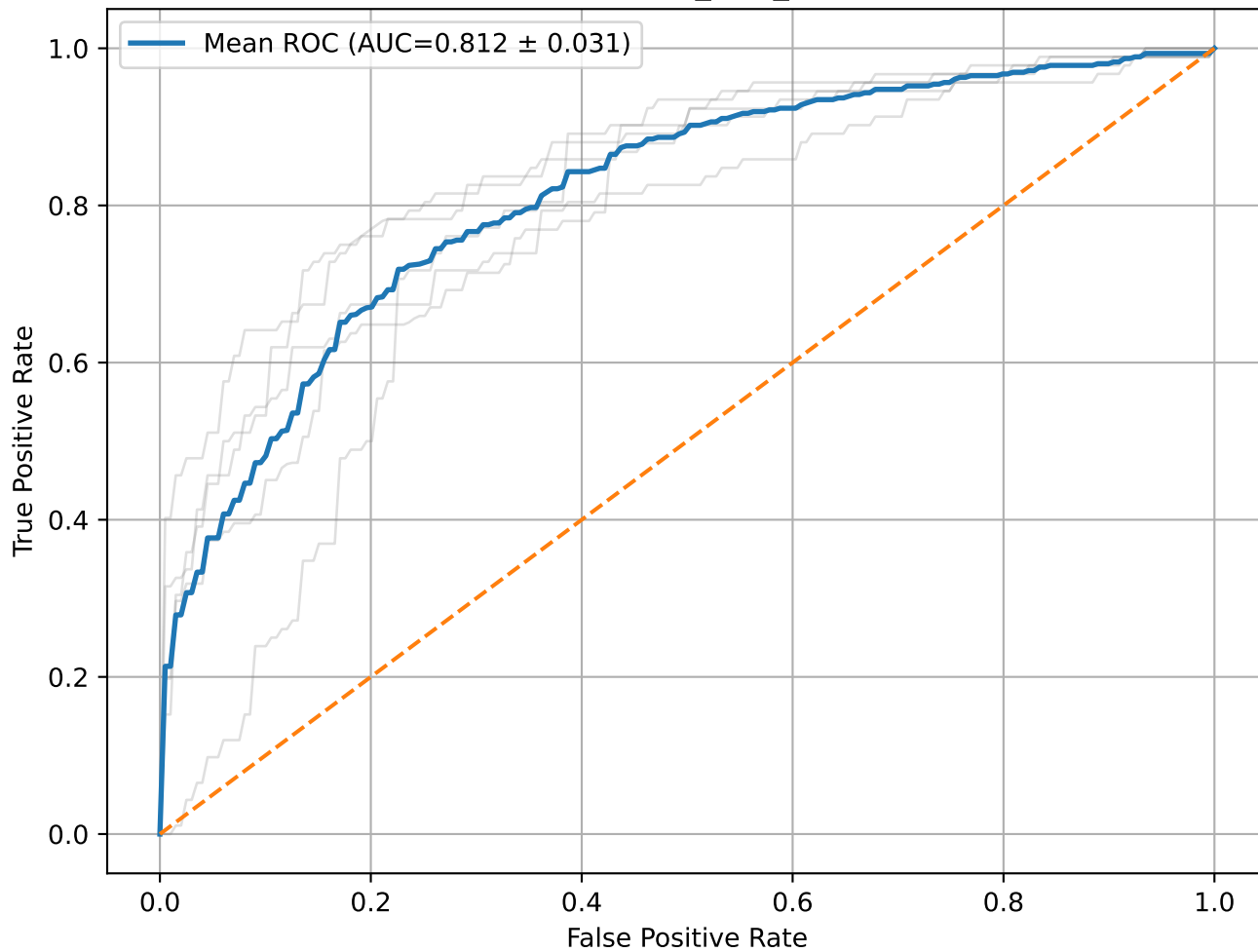
Confusion Matrix - PSO_SVM (last fold)



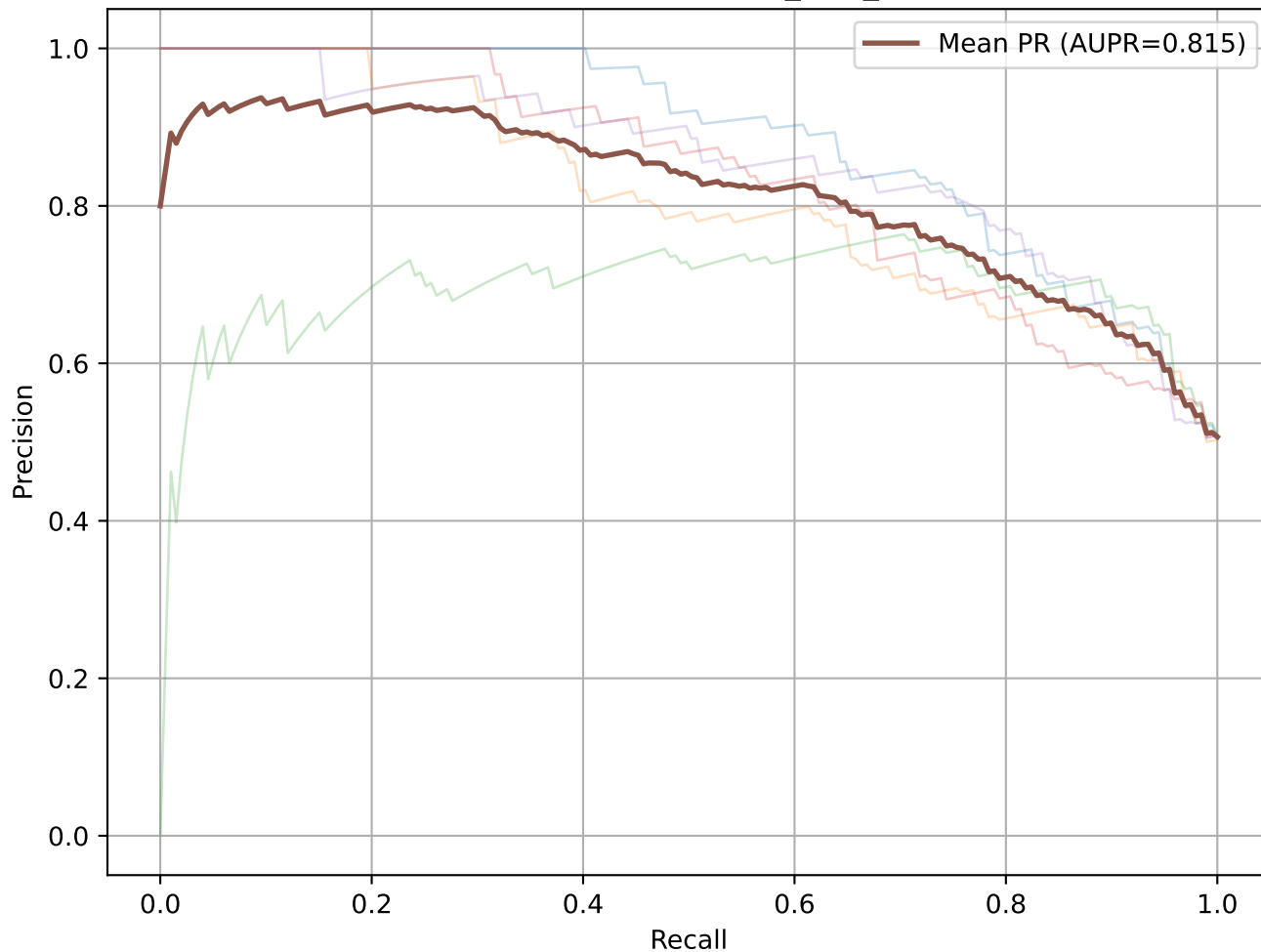
ClassificationReport - PSO_SVM (last fold)

	precision	recall	f1-score	support
0	0.85	0.82	0.83	89
1	0.83	0.86	0.84	92
accuracy			0.84	181
macro avg	0.84	0.84	0.84	181
weighted avg	0.84	0.84	0.84	181

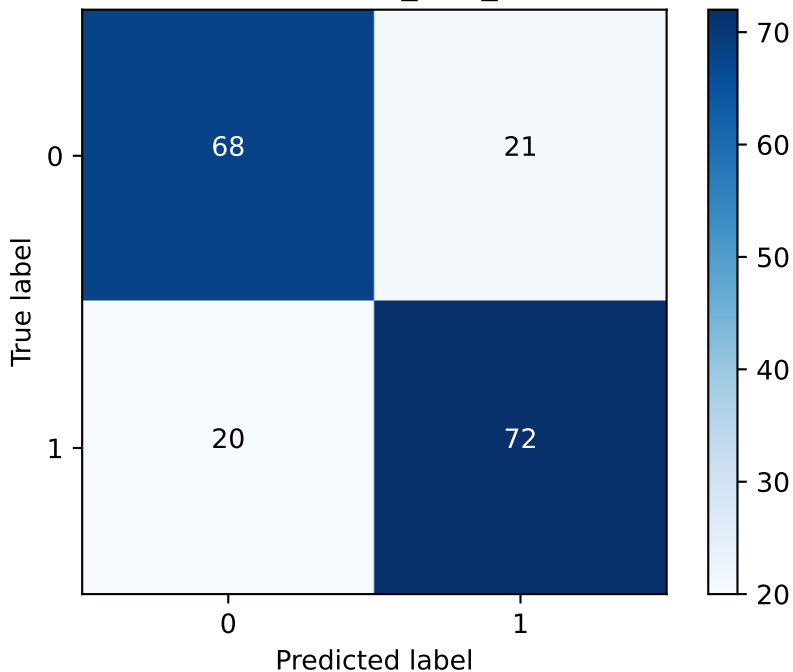
ROC - BAYES_OPT_SVMS



Precision-Recall - BAYES_OPT_SVMS



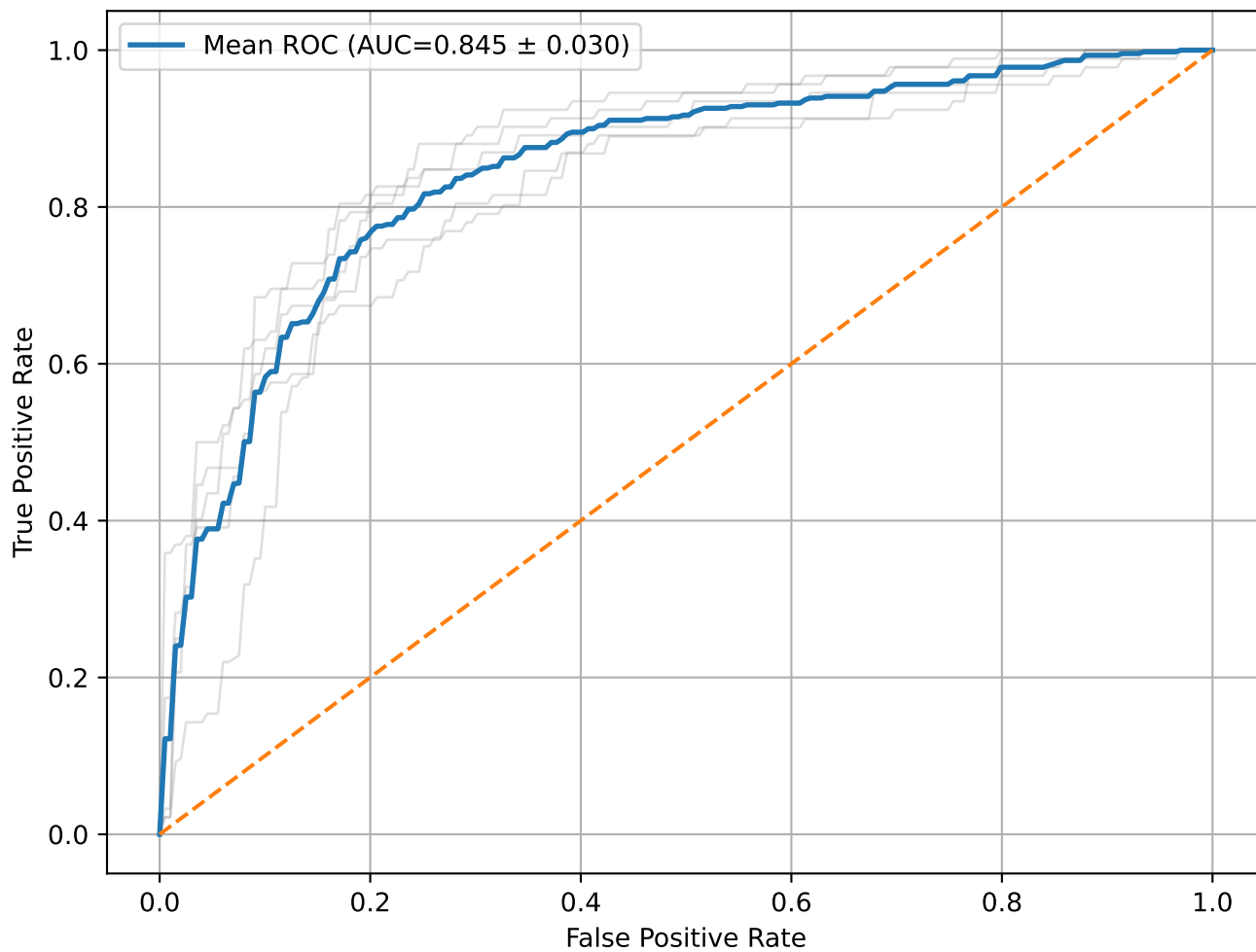
Confusion Matrix - BAYES_OPT_SVMS (last fold)



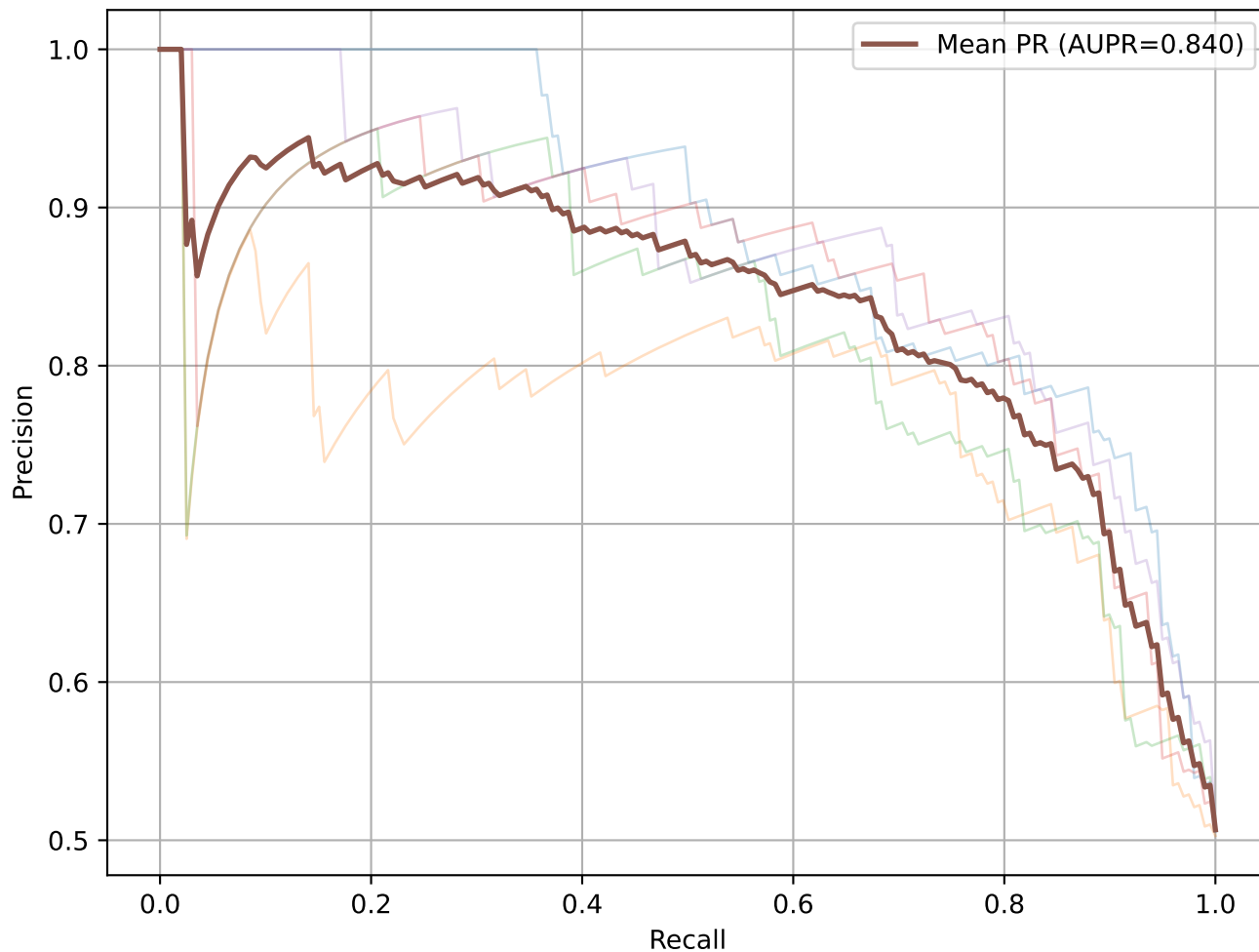
ClassificationReport - BAYES_OPT_SVMS (last fold)

	precision	recall	f1-score	support
0	0.77	0.76	0.77	89
1	0.77	0.78	0.78	92
accuracy			0.77	181
macro avg	0.77	0.77	0.77	181
weighted avg	0.77	0.77	0.77	181

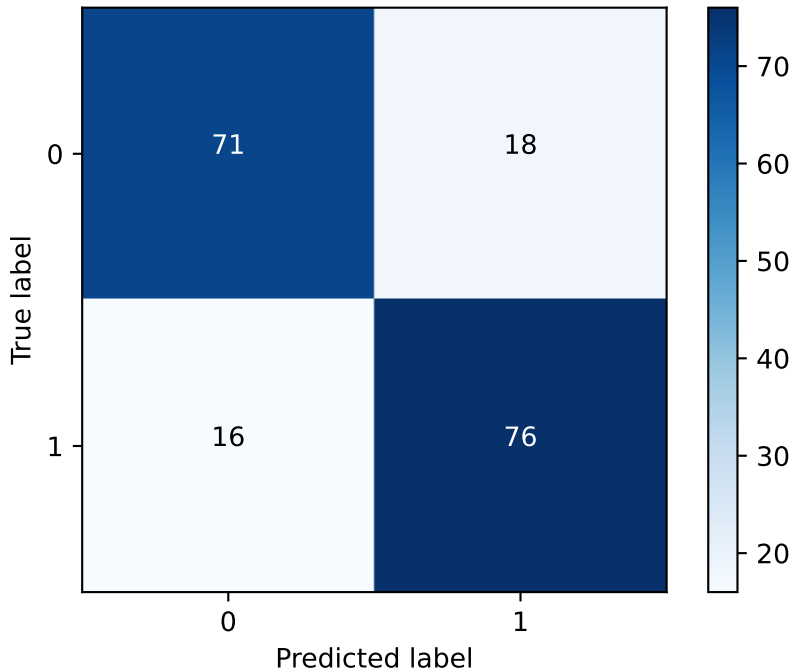
ROC - GBDT



Precision-Recall - GBDT



Confusion Matrix - GBDT (last fold)



ClassificationReport - GBDT (last fold)

	precision	recall	f1-score	support
0	0.82	0.80	0.81	89
1	0.81	0.83	0.82	92
accuracy			0.81	181
macro avg	0.81	0.81	0.81	181
weighted avg	0.81	0.81	0.81	181

Model Comparison Table (sorted by mean ROC AUC)

model	mean_roc_auc	std_roc_auc	mean_aupr	mean_accuracy	mean_f1
lgb	0.871464	0.015887	0.833885	0.801311	0.803112
voting	0.855187	0.015329	0.870045	0.775915	0.778147
stacking	0.854506	0.014717	0.869885	0.773705	0.777093
bagging	0.854133	0.020081	0.861519	0.786995	0.792236
ga_xgb	0.845725	0.021042	0.845724	0.772600	0.778792
gbdt	0.845498	0.030012	0.839843	0.784743	0.790036
xgb	0.838628	0.021402	0.844910	0.761557	0.764466
svm	0.830401	0.030675	0.829317	0.773717	0.775867
pso_svm	0.829256	0.043994	0.837370	0.763742	0.763283
rf	0.826440	0.014921	0.845564	0.747210	0.749098
logistic	0.822013	0.033500	0.821688	0.742796	0.747082
bayes_opt_svms	0.812970	0.030874	0.814959	0.739512	0.746815
pca_svm	0.810633	0.028718	0.816031	0.739469	0.737250
dnn	0.754107	0.039360	0.737199	0.682102	0.665670

Combined ROC - All Models

